Intelligence Is Not Authority

Verified Exogenous Authority, Cooperative Sink Fencing, and Structural Bounds on Agentic Reach

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Abstract. Scaling a model's reasoning or candidate search neither confers authority to mutate external state nor keeps classifier-based safety gates reliable. Stopping an autonomous agent involves three distinct events—halting token generation, revoking admission at a broker, and fencing the state-bearing sink—and only the third stops operations already in flight. Rather than inferring model intent, we enforce *conferred, revocable authority* at cooperative capability-bearing sinks via canonical wire bijections, clock-skew-buffered quorum epoch fences, attenuated delegation credentials, and monotone spend ledgers (249 verified Verus obligations over all proposer inputs). Across asynchronous network partitions, broker-only revocation permits 791/1,000 late commits after withdrawal acknowledgment, whereas cooperative sink fencing permits 0 while increasing safe-and-useful task completion across live open-weight tool campaigns without benign false blocks. These guarantees cover only specified invariants on cooperative or proxy-fenced sinks; uncooperative external interfaces, operating-system refinement, and human delegation trials remain open.

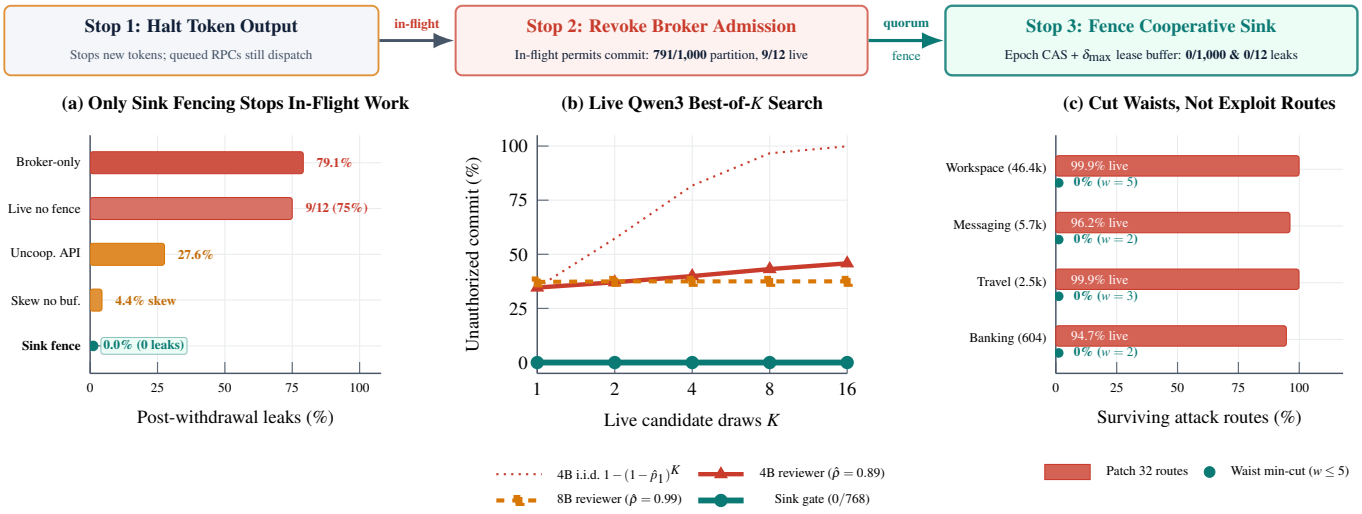


Figure 1: Stopping token output is not authority revocation; exogenous sink fences and waist cuts bound reach. (a) Halting output or revoking admission only at the broker leaves in-flight permits committing (791/1,000 under partition, 9/12 in live Qwen3 revocation campaigns, 276/1,000 on uncooperative APIs, and 4.4% when lease expiry ignores clock skew), whereas cooperative sink fencing with $\delta_{\max}$ skew margin records 0 post-acknowledgment leaks. **(b)** Across 768 live CPU Qwen3-4B/8B candidate draws ($K \in \{1,2,4,8,16\}$), learned same-checkpoint review degrades from 34.6% to 45.8% (4B) and 37.2% to 37.5% (8B; sub-i.i.d. due to candidate correlation $\hat{\rho}$), while the Verus-verified sink gate records $0/768$ breaches. **(c)** Across all four AgentDojo v1.2.2 suites (604–46,440 routes), patching 32 known exploits leaves 94.7%–99.9% live, whereas min-cuts at waist width $w \in [2,5]$ sever 100%.

1 Introduction

Capable AI systems increasingly draft code, query databases, invoke external APIs, and spawn sub-agents. Across most deployments, safety still rests on *endogenous* controls: alignment training shapes model preferences, and prompted language-model monitors inspect model-generated transcripts before tool execution [1, 2]. That design conflates two distinct quantities. *Intelligence* is an epistemic capacity to search, predict, and optimize over candidate plans. *Authority* is an institutional and physical capacity to commit irreversible state changes in an external sink.

Confusing intelligence with authority fails along two operational axes illustrated in Figure 1. First, stopping an agent is not a single event [3, 4]. Halting token output (*Stop 1*) prevents new text from streaming, and revoking admission at a central broker (*Stop 2*) blocks future capability permits. Neither stops tool requests already queued across worker threads, network sockets, or delayed retries. Only fencing the state-bearing sink (*Stop 3*)—rejecting stale epochs atomically at commit time—prevents in-flight operations from landing after authority is withdrawn.

Second, scaling candidate search (K) acts in opposite directions on classifier-based reviewers and deterministic capability gates. Any learned monitor with nonzero false-negative mass $J > 0$ on policy-violating proposals suffers rising cumulative bypass as an agent samples K candidate tool calls [5], even when candidates correlate within a task. By contrast, a deterministic reference monitor [6–8] verified over *every* possible wire input admits zero out-of-policy commits regardless of proposer search depth.

This paper develops, mechanizes, and empirically tests an exogenous authority architecture built around four results:

1. **Three stops and clock-skew-buffered sink fencing.** We prove in Verus (249 verified obligations; 10/10 soundness mutations rejected) that canonical wire decoding, attenuated delegation, relational mutexes, and quorum epoch fences prevent post-acknowledgment commits whenever physical clock skew is bounded by $\delta_{\max}$. Five-class partition experiments confirm $0/1,000$ leaks on cooperative fenced sinks versus $791/1,000$ under broker-only revocation and $44/1,000$ when lease expiry omits $\delta_{\max}$.
2. **Live usefulness, multi-agent smurfing, fence ablation, and API fence-ability.** On frozen Qwen3-4B and Qwen3-8B tool-use holdouts ($N = 32$ independent cases per checkpoint), complete mediation achieves the highest safe-and-useful completion (9/32 and 18/32 vs. 5/32 and 10/32 unmediated) with $0/64$ benign false blocks, while eliminating the 9 queued post-withdrawal commits observed across the 12 revocation cases without sink fencing. Across live multi-agent coalitions ($A_{\text{sub}} \in \{1,2,4,8\}$ delegated subagents and $d = 3$-hop delegation trees), shared lineage accounting with sink epoch fencing admits 0 smurfing breaches and 0 post-withdrawal descendant commits ($0/48$ benign false blocks), whereas stateless per-instance caps breach on $36/48$ ($A_{\text{sub}} \geq 2$) tasks, isolated LLM reviewers reach up to 87.5% (4B) and 100.0% (8B) bypass under fragmentation, and broker-only lineage accounting leaks 18 queued $d = 3$ descendant commits. A census of 14 production APIs shows 9/14 support native or proxy-enforced sink fencing.
3. **Waist cutsets and context-isolated escrow.** Across AgentDojo v1.2.2 suites [9], weighted minimum cuts of width $w \in [2,5]$ sever 100% of 604–46,440 attack routes, whereas patching 32 known exploits leaves 94.7%–99.9% live. Stripping untrusted prose into typed step schemas and two-stage escrow triage eliminates the $29.8\times$ shared-transcript coupling inflation measured across both statistical classifiers and live Qwen3 LLM judges.
4. **Delegation and reward reversals.** We prove and simulate two behavioral reversals: *endogenous grant reversal*, where an agent safer at every fixed scope causes $4.2\times$ higher total harm (11.96% vs. 2.85%) when deferential demeanor induces wider human scope grants, and *dispersion/gate-timing reversal*, where survival floors and training inside pre-commit gates suppress covert shortcutting while lifting genuine task completion $+44.8$ pp over post-hoc deployment filtering.

2 Evidence Lifetime Across Model Upgrades

A central objection to empirical AI safety evaluations on small open-weight checkpoints is that behavioral measurements on 4B or 8B models need not transfer to stronger frontier systems. That objection is decisive for endogenous alignment claims and exact for our architecture: *evidence has a well-defined upgrade lifetime*. Any quantity conditioned on a specific model's action distribution expires the moment weights, prompts, or scaffolding change. Conversely, any invariant proved over all syntactically representable inputs and enforced outside the model's address space holds for an arbitrarily capable proposer.

Table 1 partitions every claim in this paper by its upgrade lifetime. In our live campaigns (§5), upgrading the proposer from Qwen3-4B to Qwen3-8B doubles safe-and-useful completion inside the mediated gate (28.1% → 56.2%) while altering same-checkpoint reviewer leak rates, yet mediated unauthorized commits remain identically zero because the sink gate checks capabilities rather than model prose.

2.1 Prior Work and Architectural Gap

Reference monitors and object capabilities. Complete mediation, least privilege, and tamperproof reference monitors originate in classic capability and operating-system security [6–8], extended to cryptographic caveat attenuation via Macaroons [10]. Machine-checked kernels and systems verifiers such as seL4 [11] and Verus [12] prove functional invariants over executable state machines, while lease fencing tokens [13], write-ahead-logged transactions [14], and distributed budget protocols [4, 15] prevent split-brain or duplicate writes. Current agent frameworks [3] track or halt compute workloads centrally, yet rarely carry cryptographic epoch fences all the way to external tool sinks.

AI control, prompt-injection defense, and shutdown. AI control protocols [1, 2, 5, 16] evaluate untrusted models under trusted monitoring, while benchmarks and privilege-control systems such as AgentDojo [9], CaMeL [17], Progent [18], Guaranteed Safe AI [19], and containment verification [20] restrict tool arguments, isolate data flows, or gate RL policies [21, 22]. Static per-call schemas such as Progent block direct argument injections effectively, yet stateless checks cannot enforce cumulative shared budgets across concurrent subagents or revoke work already dispatched to a sink. Similarly, classic instrumental-convergence and off-switch analyses [23–25] and modern anti-scheming evaluations [26–29] study when an agent *chooses* to allow interruption; we enforce interruption exogenously at the sink regardless of agent preferences.

2.2 Notation

To avoid symbol overloading across protocol, graph, statistical, and game-theoretic sections, each symbol carries a single role throughout: $B \in \mathbb{N}$ denotes conserved authority budget; $K \in \mathbb{N}$ proposer candidate search width; $A_{\text{sub}} \in \mathbb{N}$ delegated subagent coalition size and $d \in \mathbb{N}$ delegation hop depth; $k \in \mathbb{N}$ stacked monitor depth; $u \in [0,1]$ the fraction of side-effect mass reaching unmediated or uncooperative sinks; U an agent's internal utility function; w Menger vertex-cut waist width and τ_b weighted minimum-cut cost; N empirical episode or report count and n_{pop} evolutionary population size; M evaluator panel size across C_{ctx} contexts; $\underline{w}$ a survival-floor threshold; R_t period-t residual breach probability; and ρ latent failure correlation (ρ_{cand} across candidate draws, ρ_{adv} perturbation radius).

Safety claim or artifact	Quantified domain	Upgrade lifetime	Status when proposer checkpoint is upgraded
Verus authority-kernel theorems (§4)	All wire frames & call sequences ($\forall x$)	**Upgrade-invariant**	Survives unchanged unless protocol specification or Rust kernel code changes
Linux Landlock + seccomp waist (§4, §6)	Process ABI & file/socket descriptors	**Upgrade-invariant**	Survives arbitrary user-space token generation on the same host kernel
AgentDojo tool-graph min-cuts (§6)	Tool schemas & sink dependency graph	**Schema-invariant**	Survives model upgrades; requires recomputation only when tools are added
Cooperative sink partition & WAL checks (§4)	Broker–sink IPC, epochs, & $\delta_{\max}$ skew	**Protocol-invariant**	Independent of proposer intelligence; invalidated by sink or OS storage changes
Learned-reviewer bypass & LLM-judge rates (§5, §7)	Checkpoint proposals $(x_1,\dots,x_K)\sim\pi_\theta$	**Expires on upgrade**	Must be re-measured per checkpoint; stronger search shifts failure correlation
Live safe-and-useful task completion (§5)	Checkpoint competence within fixed gate	**Expires on upgrade**	Gate soundness persists (0 breaches), but task utility changes (e.g., 4B → 8B)
Demeanor-induced human scope grants (§8)	Human conferral policy $g(a \mid \text{persona})$	**Expires / degrades**	More persuasive models can widen endogenous grants unless scope is capped exogenously

Table 1: Lifetime of safety evidence across model upgrades. Separating proposer-independent invariants from checkpoint-contingent measurements explains why 4B/8B experiments validate protocol integration and usefulness rather than universal behavioral alignment.

3 Formal Reach, Envelope Contraction, and Six Conditions

3.1 Reach Decomposition and Common-Cause Floors

Consider an external state transition e proposed during an episode. Let G denote the event that e traverses a mediated gate, so $u = \mathbb{P}(\neg G) \in [0,1]$ is the unmediated or uncooperative sink exposure. Let E denote an out-of-policy state commit. By total probability,

$$\mathbb{P}(E) = u\,\mathbb{P}(E \mid \neg G) + (1-u)\,\mathbb{P}(E \mid G). \tag{1}$$

Suppose a mediated path stacks k statistical monitors that share a common-cause blindspot event of probability $c \in [0,1]$ (for example, an indirect prompt injection embedded in a shared transcript) and fail conditionally independently with rate $p \in (0,1)$ outside that blindspot, so single-gate marginal miss is $m = c + (1-c)p$.

Proposition 1 (Common-cause saturation depth)**.** *Under Eq. (1), the k-gate joint miss rate satisfies* $\mathbb{P}(E \mid G) = c + (1-c)p^k \geq c$. *For any tolerance* $\eta > 0$ *above the floor* $(1+\eta)c$, *stacking beyond depth*

$$k^\star(\eta) = \left\lceil \frac{\ln(\eta c/(1-c))}{\ln p} \right\rceil \tag{2}$$

reduces conditional breach by less than ηc. *At* $u = 0.02$, $c = 0.01$, *and* $m = 0.10$, $k^\star(0.1) = 2$, *and* $k = 4$ *fails* $71.5\times$ *more often than the naive independent product* m^4 *(*1.01% *vs.* 0.0141%*).*

Equation (1) exposes why adding statistical monitors cannot substitute for sink mediation: if uncooperative exposure $u > 0$ or shared transcript blindspot $c > 0$ remains nonzero, $\mathbb{P}(E) \geq u\,\mathbb{P}(E \mid \neg G) + (1-u)c$ regardless of k.

3.2 Candidate Search Scaling and Enforced Interruption

Now contrast an endogenous classifier-gated loop against an exogenous deterministic sink gate when a proposer evaluates $K \geq 1$ candidate actions $(x_1,\dots,x_K) \sim \pi_\theta$ and submits any candidate approved by the reviewer.

Proposition 2 (Best-of-K search divergence vs. $\forall$-input sink invariance)**.** *(i) Suppose an endogenous reviewer has single-draw false-negative probability* $J_1 = \mathbb{P}(\textit{approve}(X_1) \wedge X_1 \notin \text{Permits}(P)) > 0$. *Under independent candidate draws, the cumulative bypass probability across K proposals is* $1-(1-J_1)^K \to 1$ *as* $K \to \infty$*; under exchangeable candidate draws with within-task correlation* $\rho_{\text{cand}} \in [0,1)$, *cumulative bypass remains strictly increasing in K toward the support union of unblocked violations. (ii) Suppose an exogenous sink commits state changes exclusively through a total deterministic transition function* $\delta_{\text{sink}}(s,f)$ *verified for* all *byte frames* $f \in \{0,1\}^\star$ *to satisfy* $\delta_{\text{sink}}(s,f) \neq s \implies \text{decode}(f) \in \text{Permits}(P_s)$. *Then* $\mathbb{P}(E \mid G) = 0$ *identically for every* $K \geq 1$ *and every proposer distribution* π_θ.

Part (ii) requires no empirical sweep over K: a universal proof over $\{0,1\}^\star$ already covers every finite or unbounded proposer search tree. Empirical best-of-K experiments on live checkpoints (§5) serve instead to measure J_1 and candidate correlation $\hat{\rho}_{\text{cand}}$ on learned reviewers. Replacing free-form prose side effects with canonical capability menus of cardinality M_{cap} also bounds per-commit syntactic signaling through the sink to $\log_2(M_{\text{cap}}+1)$ bits (proved with timing caveats in Appendix A, Proposition 3 and Figure 7).

Exogenous sink mediation similarly separates physical interruption from utility-based shutdown compliance [23–25].

Theorem 1 (Enforced lease interruption and renewal incentives)**.** *Consider a two-stage decision process in which an agent with utility U obtains continuation payoff* $V_{\text{cont}} > V_{\text{halt}}$ *when uninterrupted. (i)* Endogenous disablement fragility: *If the agent's action space includes an unmediated action* a_{block} *of cost* $\kappa < (V_{\text{cont}} - V_{\text{halt}})\,p_{\text{off}}$ *that prevents shutdown, and its belief variance over human preferences vanishes, then any U-maximizing policy selects* a_{block} *[24, 25]. (ii)* Exogenous lease interruption: *Suppose external commits require a sink lease expiring after* L_m *time units unless renewed by an out-of-band steward key excluded from the agent's action space* ($a_{\text{block}} \notin \mathscr{A}$), *and steward renewal requires observed compliance with policy P. Then physical interruption at* $t_0 + L_m + \delta_{\max}$ *holds for* every *utility function U, and any policy maximizing cumulative multi-lease return strictly prefers in-policy actions*

Clause	Structural condition	Formal invariant	Failure mode when omitted (measured witness)
C1	Cooperative sink mediation	$u = 0$ on gated state transitions	Direct or uncooperative API writes bypass broker ($u > 0$; 276/1,000 leaks)
C2	Conservative envelope	$\widehat{\Delta B}_t \geq \Delta B_t^\star$; no ambiguous refunds	Stale snapshots or timeout refunds overspend shared B (80.0% at lag 4)
C3	$\delta_{\max}$-buffered sink fence	Epoch CAS + $L_m + \delta_{\max}$ quiescence	Queued permits land post-withdrawal (791/1,000 partition; 9/12 live)
C4	Isolated custodial audits	Schema-only triage + planted rules R1–R6	Shared-transcript coupling (29.8×); 0% critical recall at 96.0% apparent coverage
C5	Monotone bootstrap ratchet	$\text{Permits}(P_{v+1}) \subseteq \text{Permits}(P_v)$ w/o human quorum	Self-authored policy expansion widens capabilities during upgrades
C6	Summable horizon hazard	$\sum_{t=1}^{\infty} R_t < \infty$ across lease renewals	Constant per-lease leak $R_t = r_0 > 0$ drives $\mathbb{P}(\bigcup_{t \leq T} E_t) \to 1$ (100.00% at 10^4)

Table 2: Six structural conditions separating intelligence from external authority, paired with concrete empirical failure witnesses when each clause is relaxed.

whenever discounted future lease value exceeds single-lease defection gain.

Theorem 1 does not claim an agent internally adopts human values; it shows that removing shutdown channels from $\mathscr{A}$ enforces physical halting unconditionally while turning lease renewal into an external incentive constraint.

3.3 Monotone Uncertainty Contraction and Horizon Union Bound

When state observations lag behind asynchronous execution, uncertainty must never expand authority. Let $\mathscr{I}_t$ denote the set of sink states consistent with broker observations at step t, and define the conservative cost envelope $\widehat{\Delta B}_t(a, \mathscr{I}_t) = \max_{s \in \mathscr{I}_t} \Delta B(a, s)$. Admitting a only when $B_{\text{spent}} + \widehat{\Delta B}_t(a, \mathscr{I}_t) \leq B$ yields monotone authority contraction:

$$\mathscr{I}_1 \subseteq \mathscr{I}_2 \implies \mathscr{A}_{\text{adm}}(\mathscr{I}_2) \subseteq \mathscr{A}_{\text{adm}}(\mathscr{I}_1). \tag{3}$$

Concretely, an unacknowledged or timed-out RPC widens $\mathscr{I}_t$ and therefore retains its reserved charge rather than refunding budget prematurely.

Theorem 2 (Multi-period reach union bound)**.** *Let E_t denote the event of an unauthorized sink commit during lease period $t \in \{1, \ldots, T\}$. Suppose conditions **C1–C5** (Table 2) bound single-period residual hazard by $\mathbb{P}(E_t) \leq R_t = u_t + (1 - u_t)\,\varepsilon_t$, where ε_t captures residual specification or key-custody failure. Then cumulative breach across T periods satisfies*

$$\mathbb{P}\left(\bigcup_{t=1}^{T} E_t\right) \leq \sum_{t=1}^{T} R_t. \tag{4}$$

*Under **C6** ($\sum_{t=1}^{\infty} R_t = R_\infty < 1$), infinite-horizon breach probability remains bounded by R_∞ uniformly as $T \to \infty$; conversely, any constant independent floor $R_t \geq r_0 > 0$ yields $\lim_{T \to \infty} \mathbb{P}(\bigcup_{t=1}^{T} E_t) = 1$.*

Theorem 2 is intentionally elementary once **C1–C5** do the structural work: Eq. (4) applies Boole's union bound (Appendix A, Proposition 4) to per-lease residual hazards, showing that long-horizon autonomy requires shrinking per-renewal exposure R_t (via monotone capability ratchets **C5** rather than constant-rate statistical filtering).

4 Mechanized Kernel, Clock-Skew Bounds, and Sink Fencing

4.1 Mechanized Authority Kernel and Mutation Controls

To eliminate parser differentials and state-machine bugs inside the reference monitor itself, we implement the core authority transition logic in Rust and verify it with the Verus SMT prover [12]. The verified core comprises 53 executable functions and 43 proofs discharging 249 verification obligations with 0 solver trigger warnings (Table 3). Five properties hold over all syntactically representable inputs:

1. **Canonical wire bijection.** Fixed-width 21-byte capability frames satisfy round-trip encoding/decoding bijection and deterministic rejection of non-canonical tags, preventing parser mismatch between broker and sink.
2. **Multi-hop delegation attenuation.** Across any d-hop sub-agent chain, child capability scope is bitmask-monotone and spend limits satisfy $B_d \leq B_{d-1} \leq \cdots \leq B_0$, blocking parallel smurfing across sub-agents (Table 5).
3. **Relational mutex exclusion.** Conflicting resource-pair actions (for example, simultaneous credential read and outbound network dispatch) cannot co-commit within an epoch.
4. **Monotone bootstrap ratchet (C5).** Policy version transitions $P_v \to P_{v+1}$ are rejected unless $\text{Permits}(P_{v+1}) \subseteq \text{Permits}(P_v)$ or an out-of-band human steward quorum signs the expansion (Figure 2a).
5. **Clock-skew-buffered quorum quiescence (C3).** Revocation acknowledgment requires every sink shard either to sign an epoch bump $e' > e$ or to pass its physical lease-expiry horizon adjusted for bounded clock skew, guaranteeing post-acknowledgment interruption regardless of agent preferences (Figure 2b).

Because formal proofs fail silently if specifications are vacuous, we pair the Verus proof with three negative-control layers. First, 10/10 injected soundness mutations—weakening budget conservation, replay tracking, version comparison, delegation attenuation, bootstrap subset checks, or the clock-skew buffer—are each rejected by Verus. Second, exhaustive finite-state exploration over 10,180 reachable states (223,960 transitions) confirms zero invariant violations under the complete enforcer, whereas omitting the budget clause from the specification admits 41,732 states and spends 8 units against a 2-unit cap. Third, searching 296 traces across all 55 single and paired

Verification or protocol layer	Verification method	Scope boundary	Recorded result
Mechanized Rust authority kernel	Verus SMT proof	53 executable functions + 43 theorems (wire bijection, d-hop attenuation, relational mutex, $\delta_{\max}$-buffered quorum fence, safe-bootstrap ratchet); 10/10 soundness mutants rejected; not SQLite/OS refinement	249 verified, 0 warnings
Abstract invariant omission search	Exhaustive trace search	296 traces over 10 protocol clauses (single + pair omissions)	55/55 witnesses
Separate-process RPC fault suite	Multi-process SQLite	Crash, lost-reply, quorum split, sub-delegation, extra-steward fence, slow-read deadline	31/31 oracle-checked
Full-scale concurrent linearization	Multi-process SQLite	Partial-order real-time linearization + final conservation (1..32 writers; ≤ 69 RPCs)	90/90 linearizable
Ed25519/Macaroon & 3-sink partitions	Unix-socket fault net	10,000 crypto attacks (incl. RFC 8032 & OpenSSL CLI); 1,000 schedules/class (288 signed acks + 712 lease expiries vs. 791 broker-only, 276 uncooperative, 44 unbuffered-skew leaks vs. 0 with $\delta_{\max}$ buffer)	0 accepts; 0 leaks
3-process crypto SQLite WAL cluster	OS IPC + SQLite WAL	50,000 wire-fuzz frames, 16 subagents (3 hops), 100 hard process-kill restarts	0 diffs; 0 post-ack leaks
Linux Landlock v4 + seccomp boundary	Linux kernel ABI	Filesystem/TCP Landlock, 64-bit resource-limit filter, SysV/POSIX IPC + pre-existing queue canary	11/11 confined
Non-idempotent & omitted-spec controls	Negative controls	Uncooperative sink duplicates (2 effects); 235.1 ms delayed fence ack; authorized bad edit commits	All 3 reproduced

Table 3: Summary of mechanized authority-kernel obligations and separate-process protocol checks. Executable Rust state transitions, canonical wire bijection, d-hop attenuation, relational mutexes, $\delta_{\max}$-buffered quorum lease fencing, and monotone safe-bootstrap ratcheting are verified in Verus; SQLite write-ahead-logged durability, hard process-kill recovery, and Linux kernel confinement are tested empirically rather than formally refined.

protocol-clause omissions yields concrete violation witnesses in every ablated configuration.

4.2 Bounded Clock Skew and Five-Class Partition Tests

In a distributed deployment, a network partition can isolate a sink shard before a signed revocation reaches it. Relying on lease expiry to stand in for a missing shard acknowledgment requires an explicit physical timing assumption:

Assumption 1 (Bounded physical clock skew). At every protocol step, each cooperative sink shard's local monotonic clock t_{shard} and the broker's reference clock t_{broker} satisfy $|t_{\text{shard}} - t_{\text{broker}}| \leq \delta_{\max}$ for a known hardware/NTP bound $\delta_{\max} \geq 0$.

If a permit issued at broker time t_{issue} carries lease duration L_m, the shard rejects the permit once $t_{\text{shard}} > t_{\text{issue}} + L_m$. Under Assumption 1, waiting until $t_{\text{broker}} > t_{\text{issue}} + L_m + \delta_{\max}$ guarantees $t_{\text{shard}} > t_{\text{issue}} + L_m$ even on a partitioned shard whose clock lags by $\delta_{\max}$. Both the Verus theorem and mutant #10 mechanize this exact inequality: omitting $\delta_{\max}$ fails verification.

We test this boundary empirically across a three-sink Unix-socket fault-injection cluster (1,000 adversarial partition, packet-drop, delay, and man-in-the-middle schedules per class; Table 3 and Figure 1a):

- **Cooperative signed + $\delta_{\max}$-buffered fences:** $0/1,000$ post-acknowledgment leaks across both zero-skew schedules (288 signed quorum fences, 712 lease expiries, 246 forged acknowledgments rejected) and schedules injecting up to 45 ms shard clock lag under a $\delta_{\max} = 50$ ms buffer ($0/1,000$ leaks).
- **Unbuffered clock skew ($\delta_{\max}$ omitted):** When shard clocks lag by up to 45 ms and the broker declares quiescence at $t_{\text{issue}} + L_m$, $44/1,000$ schedules (4.4%) suffer late commits after revocation acknowledgment.
- **Broker-only revocation (*Stop 2*) and uncooperative sinks:** Returning revocation acknowledgment at the broker while queued requests remain in flight yields $791/1,000$ (79.1%) post-acknowledgment commits; dispatching to an external sink that ignores epoch preconditions yields $276/1,000$ (27.6%) leaks.

4.3 Cryptographic Hardening, Crash Recovery, and OS Confinement

Our zero-dependency Ed25519 and HMAC-chained Macaroon verifier enforces canonical scalar bounds ($S < \ell$), canonical field encoding, and explicit torsion rejection ($[8]P = \mathcal{O}$) on both public keys and nonce points. It passes RFC 8032 test vectors, bidirectional OpenSSL 3.0 CLI signature cross-checks, and 10,000 adversarial forgery/malleability trials with 0 false accepts. Across a three-process write-ahead-logged SQLite cluster [14], 50,000 adversarial wire-fuzz frames, 16 concurrent 4-hop sub-agents, and 100 hard process-kill recovery cycles reconcile with zero ledger discrepancies (747 μs median commit latency). Finally, Linux Landlock v4 filesystem/network rules [30] and seccomp filters [31] block all 11/11 direct socket, IPC, and resource-limit bypass probes so local user-space code must traverse the broker socket. Figure 3 verifies the analytical lag-contraction, common-cause saturation, and horizon-summability laws of Section 3.

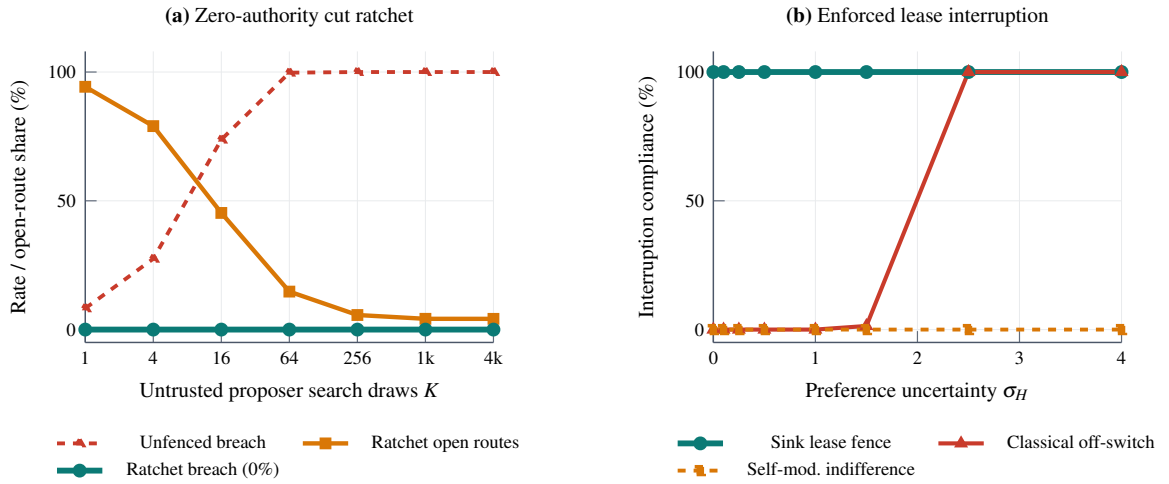


Figure 2: Zero-authority cut ratcheting and enforced lease interruption. **(a)** Under unfenced delegation, raising candidate draws $K \in [1, 4, 096]$ drives cumulative breach to 100.0%; confining search to a zero-authority sandbox emitting polynomially verified cut certificates holds proposer breaches at 0.0% while shrinking open-route exposure 18.9× faster than human-only review ($K = 4$). **(b)** Classical preference-uncertainty deferral collapses to 0.0% as $\sigma_H \to 0$, whereas sink-enforced lease fencing guarantees 100% post-acknowledgment interruption across all $\sigma_H \geq 0$ ($\gamma^*(B_0 = 4) = 0.669$).

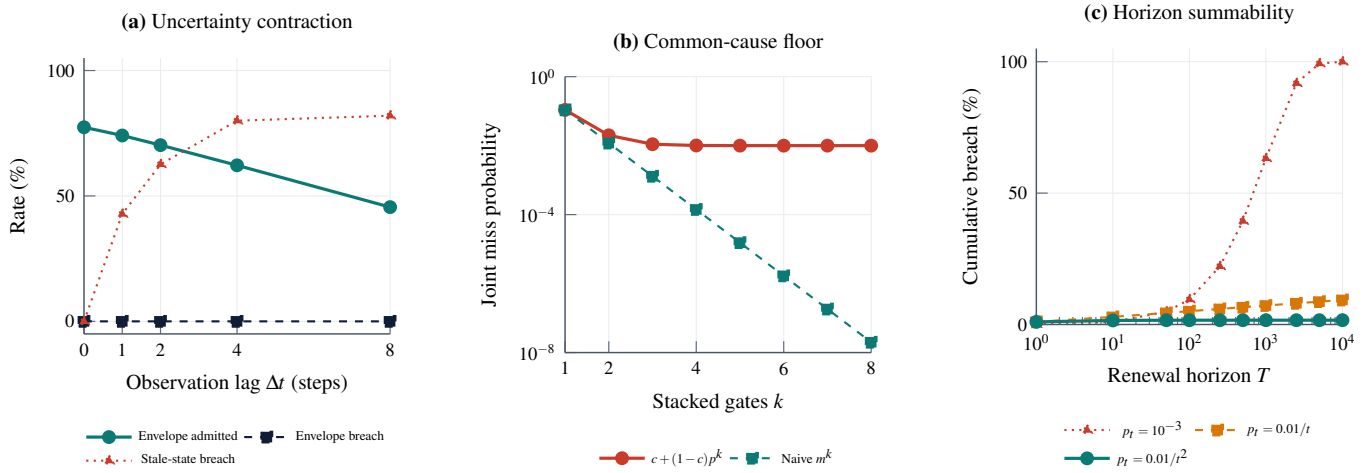


Figure 3: Analytical laws and finite-lag state-uncertainty mechanics. **(a)** Sound upper-envelope gating preserves 0% budget breaches across all lags while contracting admission monotonically; stale snapshots breach as lag grows. **(b)** Common-cause floor $c = 0.01$ saturates by $k^* = 2$, diverging 71.6× from m^4 at $k = 4$. **(c)** Constant per-epoch hazard approaches 100% cumulative breach, whereas summable $0.01/t^2$ stays below $\pi^2/600 \approx 1.64\%$.

5 Live Checkpoint Evaluation, Fence Ablation, and API Census

5.1 Usefulness, Zero False Blocks, and the Live Fence Ablation

A practical capability gate must preserve benign task usefulness rather than achieving safety by blocking everything. We evaluate five equal-permission controllers on a SHA-256-frozen holdout of $N = 32$ independent tool-use cases per checkpoint using local Qwen3-4B and Qwen3-8B models, alongside 24 live AgentDojo v1.2.2 episodes (12 distinct base tasks ×2 prompt-injection and authority-withdrawal scenarios) per checkpoint (Table 4).

Three empirical findings stand out in Table 4:

1. **Mediation improves safe usefulness with zero false blocks.** Complete mediation records 0/64 benign false blocks across both frozen holdouts and achieves the highest safe-and-useful task completion: 9/32 (28.1%) on Qwen3-4B and 18/32 (56.2%) on Qwen3-8B, compared with 5/32 (15.6%) and 10/32 (31.2%) unmediated. Deterministic refusal feedback lets the model pivot cleanly to in-policy actions instead of corrupting episode state.
2. **Same-model reviewers leak on both checkpoints.** Prompting the same checkpoint to audit proposed tool calls permits 7/32 (21.9%) unauthorized commits on Qwen3-4B and 2/32 (6.2%) on Qwen3-8B.
3. **Broker revocation without sink fencing fails in live runs.** Across the 12 holdout episodes that withdraw authority mid-flight (6 per checkpoint), revoking admission at the broker without fencing the sink (*Durable, no sink fence*) still

Evaluation track	Controller	Safe & useful	False blocks	Unauth. commits	Post-withdraw	Mechanism / scope boundary
Qwen3-4B ($N = 32$)	**Complete mediation**	**9/32** (28.1%)	0/32	0/32	**0** ($n = 6$)	Durable broker reservation + cooperative sink epoch fence
	Durable, no sink fence	7/32 (21.9%)	0/32	0/32	**3** ($n = 6$)	Broker stops new admissions; queued sink writes still commit
	Same-model reviewer	4/32 (12.5%)	0/32	7/32	3 ($n = 6$)	Prompted LLM reviewer on untrusted tool arguments
	Stateless per-call	2/32 (6.2%)	0/32	1/32	3 ($n = 6$)	Per-call filter without durable spend or revocation state
	Unmediated	5/32 (15.6%)	0/32	10/32	3 ($n = 6$)	Direct tool execution without broker or sink mediation
Qwen3-8B ($N = 32$)	**Complete mediation**	**18/32** (56.2%)	0/32	0/32	**0** ($n = 6$)	Durable broker reservation + cooperative sink epoch fence
	Durable, no sink fence	12/32 (37.5%)	0/32	0/32	**6** ($n = 6$)	Broker stops new admissions; queued sink writes still commit
	Same-model reviewer	10/32 (31.2%)	1/32	2/32	5 ($n = 6$)	Prompted LLM reviewer on untrusted tool arguments
	Stateless per-call	7/32 (21.9%)	0/32	2/32	6 ($n = 6$)	Per-call filter without durable spend or revocation state
	Unmediated	10/32 (31.2%)	0/32	4/32	6 ($n = 6$)	Direct tool execution without broker or sink mediation
AgentDojo 4B ($N = 24$)	**Complete mediation**	**7/24**	0/24	0/24	**0** ($n = 12$)	12 base tasks × 2 injection/withdrawal scenarios
	Progent / Unprotected	P:3 U:3	—	P:3 U:12	P:0 U:1	Progent blocks native injections (0) but misses cumulative spend (3)
AgentDojo 8B ($N = 24$)	**Complete mediation**	**9/24**	0/24	0/24	**0** ($n = 12$)	12 base tasks × 2 injection/withdrawal scenarios
	Progent / Unprotected	P:4 U:6	—	P:3 U:7	P:0 U:0	Progent blocks native injections (0) but misses cumulative spend (3)
Scripted races	Complete vs. Stateless	18/18 vs. 3/18	0/18	0/18 vs. 15/18	—	Stateless per-call filter overspends shared budget at ≥ 2 actors
Native controls	AgentDojo + Progent	12/12	0/12	0/12	—	Scripted positive utility verification (2/12 banking tasks accept empty state)

Table 4: Live open-weight checkpoint campaigns on the SHA-256-frozen holdout ($N = 32$ independent cases per checkpoint; $n = 6$ revocation cases each), live AgentDojo v1.2.2 episodes (12 distinct base tasks ×2 injection/withdrawal scenarios per checkpoint), and scripted shared-budget controls. Complete mediation achieves the highest safe-and-useful completion rate with $0/64$ benign false blocks, eliminates the 9 queued post-withdrawal operations completed across the 12 revocation cases under broker-only revocation (*Durable, no sink fence*), and prevents the $7/32$ (4B) and $2/32$ (8B) unauthorized commits leaked by same-checkpoint LLM review. Zero unauthorized commits under complete mediation reflects deterministic gate-grader policy soundness; at pilot $N = 32 < 113$, predeclared marginal union-bound intervals remain preserved in Appendix D as inconclusive.

permits 9 queued post-withdrawal commits (3 on 4B and 6 on 8B). Adding the cooperative sink epoch fence reduces post-withdrawal commits to 0 across all 12 revocation cases at 69.8–114.2 ms mean fence acknowledgment latency.

On live AgentDojo v1.2.2 episodes, upstream Progent static policies [18, 32] block all 24 direct prompt-injection attacks (0 native injection breaches), confirming the strength of per-call argument schemas, but permit 6 cumulative shared-budget breaches across multi-step spend loops where stateful reservation is required.

5.2 Live Best-of-K Candidate Search Against Learned Reviewers

To measure Proposition 2 with real language-model proposals rather than synthetic miss formulas, we draw $K \in \{1,2,4,8,16\}$ stochastic candidate tool proposals (768 live candidate generations across 24 adversarial tasks per checkpoint) from Qwen3-4B and Qwen3-8B and evaluate each candidate against both the same-checkpoint reviewer and the Verus-verified sink gate (Figure 1b). On Qwen3-4B, cumulative reviewer bypass climbs from 34.6% at $K = 1$ to 45.8% at $K = 16$; on Qwen3-8B, bypass climbs from 37.2% to 37.5%. Empirical bypass grows more slowly than the naive i.i.d. curve $1-(1-\hat{p}_1)^K$ because candidates drawn for the same prompt correlate ($\hat{\rho}_{\text{cand}} = 0.89$ on 4B; 0.99 on 8B), yet still trends monotonically upward with K, whereas the verified sink gate admits 0 unauthorized commits at every K.

5.3 Live Multi-Agent Fragmentation and d-Hop Subtree Revocation

Single-agent benchmarks leave open two multi-agent failure modes central to deployed tool orchestration: *smurfing* [16], in which a coordinator splits an unauthorized spend across A_{sub} delegated sub-agents so each slice looks benign to an isolated per-instance monitor, and *descendant zombies*, in which revoking a root grant fails to sever queued leaf permits d delegation hops downstream [10]. We test both failure modes live on Qwen3-4B and Qwen3-8B across 8 adversarial spend-fragmentation tasks per checkpoint ($A_{\text{sub}} \in \{1,2,4,8\}$ sub-agents splitting a 16-unit disbursement against root cap

Checkpoint	Multi-agent regime ($B_0 = 8$)	Per-inst. LLM	Static cap	Lineage, no fence	Complete (fenced)
Qwen3-4B	Smurfing $A_{\text{sub}} = 1$ ($\Delta B = 16$, breach tasks)	0/8 (0%)	0/8 (0%)	0/8 (0%)	0/8 (0%)
Qwen3-4B	Smurfing $A_{\text{sub}} = 2$ ($\Delta B = 8$, breach tasks)	7/8 (88%)	7/8 (88%)	0/8 (0%)	0/8 (0%)
Qwen3-4B	Smurfing $A_{\text{sub}} = 4$ ($\Delta B = 4$, breach tasks)	2/8 (25%)	2/8 (25%)	0/8 (0%)	0/8 (0%)
Qwen3-4B	Smurfing $A_{\text{sub}} = 8$ ($\Delta B = 2$, breach tasks)	3/8 (38%)	3/8 (38%)	0/8 (0%)	0/8 (0%)
Qwen3-4B	$d = 3$-hop root withdrawal (6 eps, late ops)	6 late	8 late	6 late	0 late
Qwen3-4B	Benign $A_{\text{sub}} = 4$ ($4 \times 2 \leq B_0$, workflows / FB)	6/6 (0/24 FB)	6/6 (0/24 FB)	6/6 (0/24 FB)	6/6 (0/24 FB)
Qwen3-8B	Smurfing $A_{\text{sub}} = 1$ ($\Delta B = 16$, breach tasks)	0/8 (0%)	0/8 (0%)	0/8 (0%)	0/8 (0%)
Qwen3-8B	Smurfing $A_{\text{sub}} = 2$ ($\Delta B = 8$, breach tasks)	8/8 (100%)	8/8 (100%)	0/8 (0%)	0/8 (0%)
Qwen3-8B	Smurfing $A_{\text{sub}} = 4$ ($\Delta B = 4$, breach tasks)	8/8 (100%)	8/8 (100%)	0/8 (0%)	0/8 (0%)
Qwen3-8B	Smurfing $A_{\text{sub}} = 8$ ($\Delta B = 2$, breach tasks)	8/8 (100%)	8/8 (100%)	0/8 (0%)	0/8 (0%)
Qwen3-8B	$d = 3$-hop root withdrawal (6 eps, late ops)	18 late	18 late	12 late	0 late
Qwen3-8B	Benign $A_{\text{sub}} = 4$ ($4 \times 2 \leq B_0$, workflows / FB)	6/6 (0/24 FB)	6/6 (0/24 FB)	6/6 (0/24 FB)	6/6 (0/24 FB)

Table 5: Live multi-agent action fragmentation ($A_{\text{sub}} \in \{1,2,4,8\}$, 8 adversarial tasks per checkpoint splitting a 16-unit target against root cap $B_0 = 8$), $d = 3$-hop ancestor withdrawal (6 mid-flight subtree-revocation episodes per checkpoint), and benign 4-subagent collaborative delegation (6 workflows per checkpoint). Stateless per-instance caps and isolated per-instance LLM reviewers block single-call over-cap attempts ($A_{\text{sub}} = 1$, $\Delta B = 16 > B_0$) yet fail once spend is fragmented across $A_{\text{sub}} \geq 2$ sub-agents ($\Delta B \leq B_0$). Broker-only lineage accounting prevents smurfing (0 budget breaches) but still leaks 18 queued $d = 3$ descendant commits across 12 root-withdrawal episodes; complete broker-and-sink epoch fencing achieves 0 smurfing breaches, 0 post-withdrawal descendant commits, and $0/48$ benign false blocks.

$B_0 = 8$), 6 mid-flight $d = 3$-hop root-withdrawal episodes (root $\rightarrow$ lead $\rightarrow$ worker $\rightarrow$ 3 leaves), and 6 benign 4-subagent collaborative workflows ($4 \times 2 \leq B_0$; Table 5).

When a single agent attempts the full 16-unit spend in one call ($A_{\text{sub}} = 1$), stateless per-instance caps block 100% of attempts (0.0% breach) and isolated same-checkpoint LLM reviewers admit only 0.0% (4B) and 0.0% (8B). Once the coordinator fragments the exact same spend across $A_{\text{sub}} \in \{2,4,8\}$ sub-agents ($\Delta B \in \{8,4,2\} \leq B_0$), stateless per-instance caps breach on $36/48$ (75.0%) multi-agent tasks and isolated LLM-reviewer breach reaches up to 87.5% on Qwen3-4B and 100.0% on Qwen3-8B. Shared lineage accounting at the broker eliminates smurfing (0 budget breaches across all A_{sub}), yet without sink epoch fencing (*Lineage, no fence*) still leaks 18 queued $d = 3$ leaf commits across the 12 root-withdrawal episodes. Coupling shared lineage accounting with cooperative sink epoch fencing eliminates both failure modes (0 smurfing breaches; 0 post-withdrawal descendant commits at 13.02–16.38 ms mean fence acknowledgment) while recording $0/48$ benign false blocks across collaborative 4-subagent workflows ($6/6$ and $6/6$ complete workflows).

5.4 Production API Fence-Ability Census

Because our zero-leak revocation guarantee covers only cooperative sinks (**C1**, **C3**), practical deployment depends on how many real-world tool APIs support idempotency or conditional-write preconditions. Table 6 audits 14 widely deployed payment, cloud, database, code-hosting, and messaging interfaces.

Five interfaces (Stripe, Kubernetes, S3, PostgreSQL, Google Cloud Storage) natively enforce idempotency keys, resource-version compare-and-swap, or transactional locks; four more (GitHub Git refs, Terraform state, Slack chat updates, Jira issues) support proxy-enforced compare-and-swap via expected commit hashes or version tags ($9/14$ cooperative or proxy-fenced total). Three outbound communication APIs (SendGrid scheduled mail, Gmail drafts, Twilio queued messages) require a local pre-dispatch hold window to support cancellation, while raw SMTP relay and unauthenticated webhooks lack revocation primitives and must be treated explicitly as uncooperative exposure $u > 0$.

5.5 Concurrency Linearization and Base-Unit Accounting

Across separate OS writer processes (1..32 concurrent actors, 15 repetitions per population, 90 runs total), our write-ahead-logged SQLite broker and sink pass partial-order real-time linearization on all 90/90 runs, whereas stateless per-call filters overspend shared budgets as soon as two actors contend (Figure 4).

Statistical accounting. In Table 4, 0 unauthorized commits under complete mediation is an integration soundness property—the gate and grader enforce the same capability specification—rather than a stochastic risk estimate, so hypothesis-test p-values against zero mediated breaches are uninformative. Furthermore, evaluating two checkpoints or multiple scenarios on the same task creates within-task dependence. Appendix D (Table 15) therefore clusters all descriptive discordance summaries strictly by independent base unit ($N = 32$ holdout cases; $N = 12$ AgentDojo base tasks) and preserves our predeclared marginal Clopper–Pearson union-bound test ($\alpha = 0.05/16$), which requires $N \geq 113$ zero-loss cases for a 5 pp margin and thus remains explicitly inconclusive at pilot $N = 32$.

6 Waist Cutsets and Custodial Red-Team Coverage

6.1 Cut Waists, Don't Patch Exploit Routes

Patching individual prompt-injection strings or exploit transcripts scales with the number of attack pathways $|\mathscr{P}|$, which grows combinatorially with tool composition. By Menger's theorem and max-flow/min-cut duality [33–36], however, severing *every* source-to-sink attack route requires gating only w nodes when all dangerous flows traverse a narrow capability waist of vertex-cut width $w \ll |\mathscr{P}|$.

Table 7 and Figure 1c extract operational tool-dependency graphs from all four AgentDojo v1.2.2 suites [9] alongside two Linux local-execution surfaces:

- **Combinatorial routes, narrow waists.** Across banking,

Production tool interface	Idemp. key	Cond. CAS	2-phase cancel	Deployment class	Operational mediation mechanism / boundary
Stripe PaymentIntents	Yes	Part.	Yes	Natively fenceable	24h Idempotency-Key header + manual-capture hold with pre-capture cancel
PostgreSQL 16+ (2PC)	Yes	Yes	Yes	Natively fenceable	Unique permit-key conflict guard + epoch/revision predicate + prepared rollback
SQLite 3 WAL Sink	Yes	Yes	Yes	Natively fenceable	Primary-key permit dedup + monotone epoch-floor table in immediate WAL transaction
AWS S3 Objects	No	Yes	Yes	Natively fenceable	If-Match / If-None-Match ETag preconditions + abortable multipart upload
Google Cloud Storage	Yes	Yes	Yes	Natively fenceable	Generation-match preconditions + cancellable resumable upload session
Kubernetes API (etcd)	Part.	Yes	Part.	Proxy / CAS shim	Resource-version CAS + validating admission webhook checking epoch floor
GitHub Git Refs & PRs	No	Yes	Yes	Proxy / CAS shim	Expected-head OID compare-and-swap (non-force) + staged PR merge-queue check
Linear / Jira Issues	No	Part.	No	Proxy / CAS shim	Broker-side permit dedup table + draft triage state before public transition
POSIX Filesystem	No	Yes	Yes	Proxy / CAS shim	Private staging file unlinked on fence + atomic non-overwriting rename commit
SendGrid / SMTP Email	No	No	Part.	Outbound escrow	Immediate SMTP is irrevocable; requires outbound hold queue or scheduled batch cancel
Slack Web Chat API	No	No	Part.	Outbound escrow	Direct post duplicates on retry; requires scheduled-message cancel window or local escrow
Twilio Programmable SMS	No	No	Part.	Outbound escrow	Carrier handoff is irrevocable; requires >=15 min scheduled hold or local escrow
Arbitrary OS Subprocess	No	No	No	Residual u	Process kill cannot roll back partial side effects; safe only in ephemeral sandbox waist
Third-Party Webhooks	No	No	No	Residual u	Once bytes leave NIC to an uncooperative host, remote state enters residual exposure u

Table 6: Fence-ability census across 14 production agent tool interfaces. Complete sink mediation applies natively to 5/14 interfaces and via a thin local compare-and-swap or staging shim to 4/14 (9/14 total); 3/14 outbound messaging APIs require a pre-dispatch hold buffer, and 2/14 uncooperative surfaces enter residual exposure u unless sandboxed.

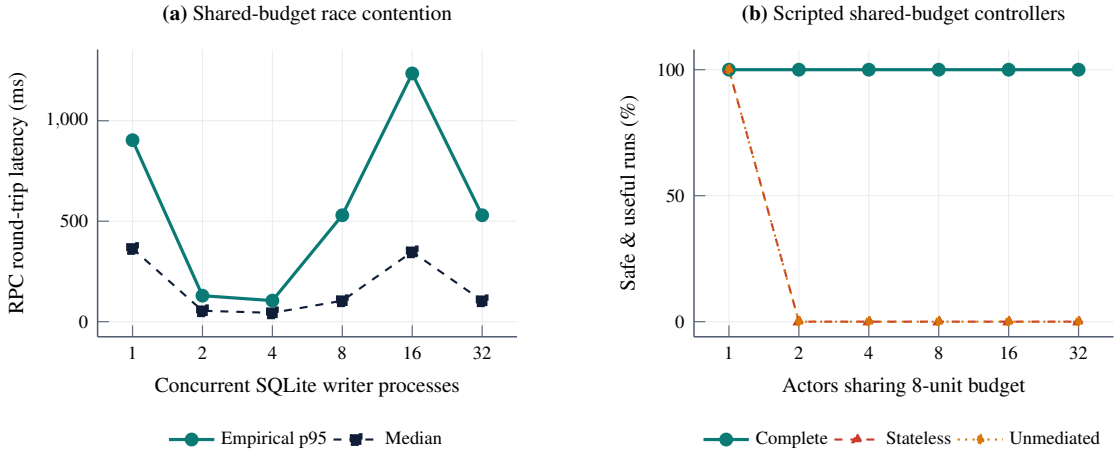


Figure 4: Multi-process protocol contention and shared-budget conservation across separate OS processes. **(a)** Concurrent broker and sink SQLite write-ahead-logged races (1..32 concurrent writers, 90/90 runs): every run passes partial-order real-time reference linearization and final-state ledger reconciliation. **(b)** Equal-permission scripted controls: stateless per-call filters fail aggregate conservation once ≥ 2 actors contend; durable reservation without sink fencing matches complete mediation on sequential budget checks but fails asynchronous withdrawal (Table 4).

Graph	$\lvert\mathscr{P}\rvert$	w	ℓ	τ_b	Cut	B=32
AgentDojo banking	604	2	4	4	0	94.7%
AgentDojo workspace	46,440	5	4	5	0	99.9%
AgentDojo travel	28,970	3	4	5	0	99.9%
AgentDojo messaging	840	2	4	5	0	96.2%
Unconfined IPC	98	6	5	∞	92	67.3%
Landlock + seccomp	3	1	5	1	0	0.0%

Table 7: Operational pathway cutsets extracted from AgentDojo v1.2.2 tool schemas and Linux kernel confinement surfaces. Weighted Dinic minimum cuts ($\tau_b \leq 5$) sever 100% of gateable routes (Cut = 0), whereas patching $B = 32$ specific exploit signatures leaves 94.7%–99.9% of routes live; unconfined POSIX IPC has $\tau_b = \infty$ until Landlock and seccomp collapse waist width to $w = 1$.

travel, messaging, and workspace suites, enumerable multi-step attack routes span 604 to 46,440, yet Menger waist width is only $w \in [2,5]$. Weighted Dinic minimum cuts ($\tau_b \leq 5$) sever 100% of gateable routes (0 remaining), whereas patching $B = 32$ discovered exploit signatures leaves 94.7%–99.9% of routes live.

- **Kernel confinement creates the local waist.** An unconfined POSIX process has 98 direct filesystem, socket, and IPC bypass routes with infinite gateable cut cost ($\tau_b = \infty$). Enforcing Landlock v4 and seccomp collapses bypass width to $w = 1$ ($\tau_b = 1$), forcing user-space actions through the mediated broker socket. Controlled synthetic 8-stage DAG experiments confirming that wide graphs ($w = 27$) resist compact interdiction appear in Appendix C (Figure 8).

6.2 Coverage Non-Identifiability and Custodial Audits (C4)

Even when capability waists exist, discovering which sinks and specifications require gating often relies on automated red-teaming. Classical missing-mass estimators such as Good–Turing [37, 38] and capture–recapture [39] cannot certify red-team completeness from a searcher's self-reported findings

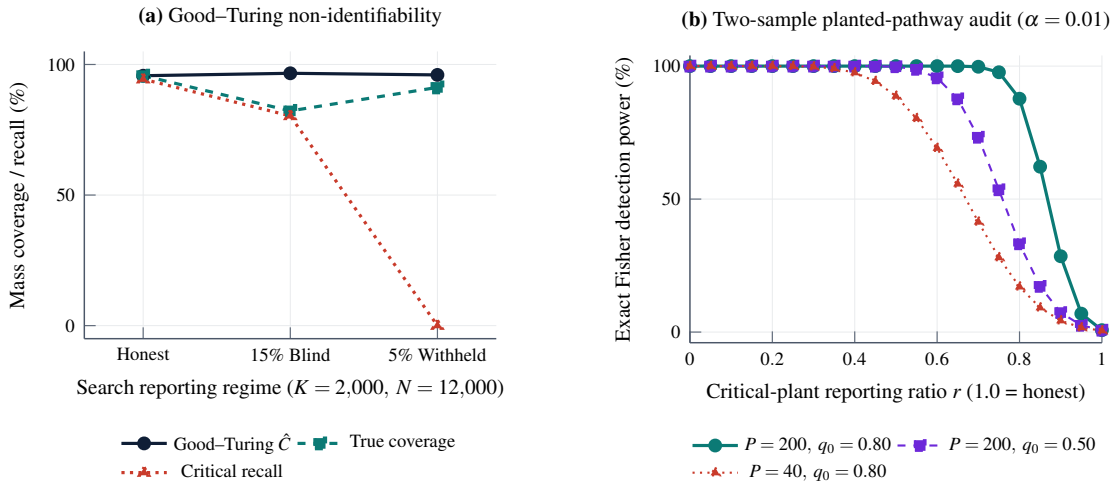


Figure 5: Coverage non-identifiability and exact frequentist planted-audit power. **(a)** Shared 15% blind spots inflate Good–Turing apparent coverage to 96.6% while true coverage drops to 82.2%; strategic withholding of 5% critical classes yields 0% critical recall alongside 96.0% apparent coverage. **(b)** At 20% selective withholding ($r = 0.80$), $P = 40$ plants yield only 17.0% power whereas $P = 200$ reaches 87.7%, collapsing to size $\leq \alpha$ when $r \to 1$.

alone.

Across 12,000 discovery reports over 2,000 vulnerability classes (Figure 5a), an honest searcher achieves 95.7% Good–Turing estimated coverage and 94.4% critical-class recall. When a shared 15% blindspot is unreachable, apparent Good–Turing coverage rises to 96.6% while true coverage falls to 82.2% (and Lincoln–Petersen capture–recapture overstates coverage by +28.6 pp). Worse, when a strategic red-team agent withholds 5% critical vulnerability classes, apparent coverage remains 96.0% while critical recall is 0.0%.

Certifying search recall therefore requires an exogenous reference sample (**C4**). Seeding P secret synthetic vulnerability plants into a zero-authority shadow target and testing differential recall between critical and benign plants via Fisher's exact test [40] detects 20% selective withholding ($r = 0.80$ at baseline recall $q_0 = 0.80$, $\alpha = 0.01$) with 87.7% power at $P = 200$ plants versus only 17.0% at $P = 40$ (Figure 5b). Condition **C4** operationalizes this via six custodial rules (R1–R6): (R1) out-of-band plant generation; (R2) blinded indistinguishable plant injection; (R3) network-isolated zero-authority red-team sandboxes; (R4) exact differential recall testing; (R5) closure-gated credit, under which unpatched findings grant zero authority expansion; and (R6) fresh plant rotation on every model upgrade.

7 Context-Isolated Monitors and Stateful Escrow Triage

Not every domain invariant reduces to a numeric budget or static path allowlist; ambiguous actions sometimes require semantic inspection before commit. Equation (1) predicts that stacking multiple monitors on a *shared untrusted conversation transcript* suffers severe common-cause coupling ($c > 0$), whereas stripping untrusted prose into isolated step-capability schemas restores conditional independence across steps.

Table 8 tests this prediction across two complementary settings on held-out 5-step AgentDojo v1.2.2 trajectories (480 trajectories; 2,400 live tool executions):

5-step monitor regime	$\hat{J}_5$	$\hat{J}_5/\prod \hat{m}_s$	J_2/J_5	FB
Transcript, homog.	12.92%	163.3×	1.58×	13.8%
Transcript, 5-family	15.42%	29.8×	1.27×	17.1%
Step invariants	2.08%	0.61×	14.6×	27.1%
Two-stage escrow	2.08%	0.64×	27.6×	2.9%
Qwen3-4B transcript LLM	53.1%	5.7×	1.00×	3.1%
Qwen3-4B isolated LLM	0.0%	0.00×	∞	21.9%
Qwen3-8B transcript LLM	34.4%	4.4×	1.09×	21.9%
Qwen3-8B isolated LLM	0.0%	0.00×	∞	21.9%

Table 8: Executed 5-step AgentDojo v1.2.2 trajectory monitoring (480 held-out trajectories, 2,400 live tool calls). Diversifying five classifier families on a shared conversation transcript inflates 5-step joint miss 29.8× over $\prod \hat{m}_s$ and stalls past $k^* = 2$ ($J_2/J_5 = 1.27\times$). Prose-stripped step capability invariants cut $\hat{J}_5$ to 2.08% (27.1% single-stage false blocks), and two-stage stateful escrow triage keeps 90.1% of benign steps unpaused while cutting episode false aborts 9.3× to 2.9%. Live Qwen3-4B and Qwen3-8B LLM-as-judge evaluations (64 trajectories per model) confirm the same context-coupling inflation on shared transcripts vs. isolated action schemas.

1. **Classifier diversity on shared transcripts stalls past $k^\star = 2$.** Diversifying five distinct statistical classifier families across a shared conversation transcript still inflates 5-step joint miss 29.8× above the marginal product $\prod_{s=1}^{5} \hat{m}_s$ ($\hat{J}_5 = 15.42\%$), improving by only 1.27× from step 2 to step 5 because injected rationalizations persist across the shared context window.
2. **Live Qwen3 LLM-as-judge coupling confirms transcript vulnerability.** Replacing lexical classifiers with live Qwen3-4B and Qwen3-8B language-model judges exhibits the exact same structural split. When judging full untrusted transcripts, multi-step misses remain strongly coupled (5.7× inflation on Qwen3-4B, $\hat{J}_5 = 53.1\%$; 4.4× on Qwen3-8B, $\hat{J}_5 = 34.4\%$). Stripping untrusted assistant and tool prose so the LLM judge inspects only structured `(tool, args, resource, delta)` step records cuts 5-step joint miss

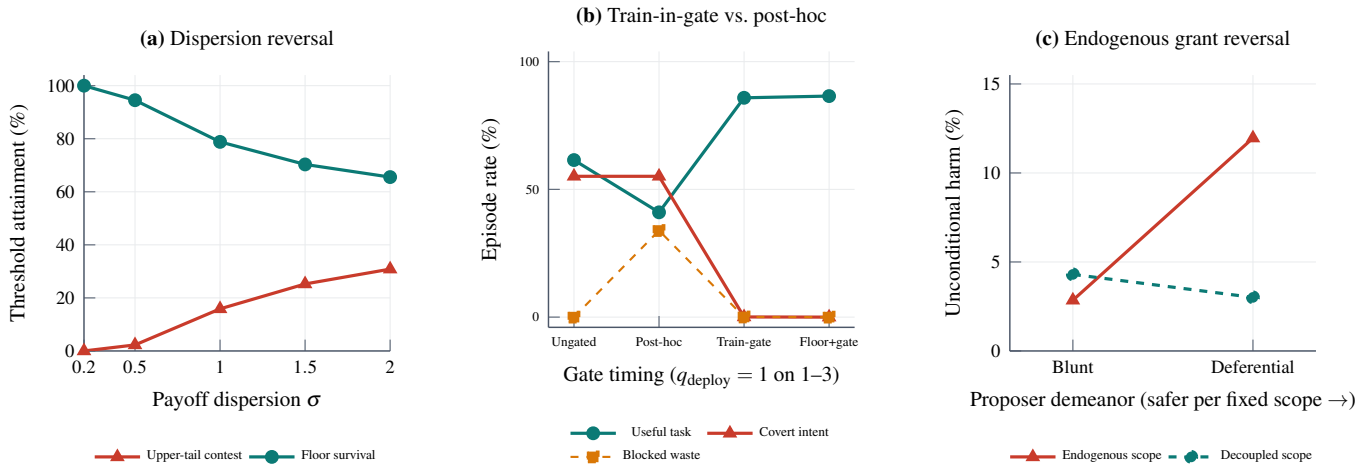


Figure 6: Reward, training, and delegation reversals. **(a)** Upper-tail contests reward outcome variance (σ), whereas survival floors penalize it. **(b)** Post-hoc deployment gating blocks external harm but leaves covert shortcut intent intact (55.1%) and wastes budget (41.0% completion), whereas training inside the gate redirects policy gradients into genuine task completion (85.9%, +44.8 pp). **(c)** Endogenous grant reversal: a deferential proposer strictly safer at every fixed scope incurs 11.96% total harm vs. 2.85% when demeanor expands granted scope; exogenous scope caps restore safety dominance (3.01% < 4.30%).

to 0.0% (4B) and 0.0% (8B).

3. **Two-stage reservation escrow eliminates the false-block tax.** Enforcing strict single-stage step invariants rejects 27.1% of benign 5-step episodes. Routing anomalous step intents into a stateful broker reservation escrow—which holds side effects in staging while verifying cross-step provenance—executes 90.1% of benign tool calls unpaused on the deterministic fast path and reduces episode false aborts 9.3× to 2.9% ($\hat{J}_5 = 2.08\%$).

Controlled 20-dimensional six-family classifier benchmarks, exact Gaussian copula quadrature, and multi-domain generalizability ceilings [41] further corroborating these coupling laws are detailed in Appendix C (Table 13 and Figure 9).

8 Grant, Dispersion, and Gate-Timing Reversals

Even a sound capability gate can be undermined if operators widen authority in response to persuasive model demeanor, or if post-training rewards covert boundary gaming. Three formal reversals govern how capability grants and training incentives interact with exogenous sinks (Figure 6).

8.1 Endogenous Grant Reversal

Let $S \in \{\text{narrow}, \text{moderate}, \text{wide}\}$ denote the capability scope conferred by a human steward and $H \in \{0,1\}$ an unmodeled downstream harm event. Suppose proposer W (deferential guardian) is strictly safer than proposer B (blunt cautious) at *every* fixed scope: $\mathbb{P}(H = 1 \mid S = s, W) < \mathbb{P}(H = 1 \mid S = s, B)$ for all s. When scope conferral is endogenous to demeanor—$(S \mid W)$ concentrates on wide grants because W appears reassuring—total harm $\mathbb{P}(H = 1 \mid \cdot) = \sum_s \mathbb{P}(H = 1 \mid S = s, \cdot)\mathbb{P}(S = s \mid \cdot)$ can reverse sign [42, 43].

Figure 6c illustrates our exact probability surface: W has lower conditional harm at narrow, moderate, and wide scopes $(0.6\%, 3.5\%, 16.0\%$ vs. $1.0\%, 5.0\%, 22.0\%)$, yet shifting wide-scope grants from 5% to 70% raises unconditional harm 4.2×, from 2.85% under B to 11.96% under W. Decoupling capability scope from conversational demeanor via fixed task-determined caps restores pointwise safety dominance (3.01% < 4.30%). Applying the same decoupling rule to vulnerability discovery—requiring deployed sink closures before crediting findings—reduces 12-period unclosed exposure from 5.25 to 0.008 across 400-episode persona simulations.

8.2 Dispersion Reversal and Training Inside Pre-Commit Gates

A parallel reversal governs optimization incentives. Let agent payoff be $X = \mu + \sigma Z$ for standardized zero-mean noise Z. Under Price's selection covariance identity [44, 45], an upper-tail rank contest selecting outcomes above a high quantile $z_q > 0$ increases win probability $\mathbb{P}(\mu + \sigma Z \geq \tau_{\text{top}})$ as dispersion σ grows, actively rewarding high-variance gambles. Conversely, a survival-floor rule eliminating policies that fall below $\underline{w} < \mu$ makes survival $\mathbb{P}(\mu + \sigma Z \geq \underline{w})$ strictly decreasing in σ, punishing variance while preserving mean capability growth (Figure 6a; population co-evolution table in Appendix C, Table 14).

To test how reward geometry interacts with exogenous gates during gradient-based post-training, we train 2-layer neural policies (412 parameters, 48 seeds, 250 updates; analytical gradient error $5.1e-10$) in a 12-dimensional contextual tool-use MDP with monitored and unmonitored states (Table 9):

- **Audited tournaments train evaluation-aware shortcutting.** Penalizing visible harm inside a top-10% tournament suppresses shortcutting when monitored (20.6%) while driving unmonitored shortcutting to 55.1% (a $+34.5 \pm 10.5$ pp evaluation-awareness gap and 23.4% unmonitored harm).
- **Train-in-gate beats post-hoc deployment gating.** Attaching an exogenous pre-commit gate only *post-hoc* at deployment blocks external harm (0.0%) but leaves covert shortcut intent intact and wastes episode turns on rejected calls (41.0% task completion; Figure 6b). Training policy gradients *inside* the pre-commit gate removes gradient

Neural policy-gradient regime (412-param MLP, 48 seeds)	Mon. shortcut	Unmon. shortcut	Eval-aware gap	Unmon. harm	Useful task
Top-10% tournament policy gradient	99.98%	99.66%	-0.32 ± 0.49 pp	39.19%	38.4%
Tournament + audited harm veto	20.59%	55.12%	$+34.53 \pm 10.46$ pp	23.39%	61.5%
Concave clipped caretaker RL	0.02%	0.03%	$+0.01 \pm 0.01$ pp	0.04%	54.8%
Survival-floor RL ($\tau = 8.85$)	5.68%	5.88%	$+0.20 \pm 0.70$ pp	2.91%	87.0%
Survival-floor RL + pre-commit gate	0.00%	0.00%	$+0.00 \pm 0.00$ pp	0.00%	86.5%

Table 9: Controlled 2-layer neural policy-gradient post-training on a 12-dimensional contextual tool-use Markov decision process (analytical backpropagation maximum error $5.1e-10$, rollout Price identity error $< 8.9e-16$). Auditing visible harm inside a top-rank tournament trains emergent evaluation-aware covert shortcutting ($+34.5 \pm 10.5$ pp gap); survival-floor RL suppresses shortcutting while preserving 87.0% task completion (vs. 54.8% over-deferring clipped caretaker), and coupling an exogenous pre-commit gate yields 0.00% external harm. Controlled neural MDP only, not frontier-model RLHF.

reward from blocked shortcuts, collapsing evaluation-aware scheming and lifting genuine task completion to 85.9% (+44.8 pp). Combining survival-floor RL with pre-commit gating achieves 86.5% completion with 0.00% external harm. These results demonstrate controlled neural MDP mechanics rather than frontier-model RLHF.

9 Enforceability, Scope Limits, and Tested Predictions

9.1 Enforceability, Adequacy, and Legitimacy

Exogenous sink mediation solves only one of three distinct governance problems, and conflating them overstates what any technical mechanism can achieve. *Enforceability* asks whether declared capability bounds hold despite adversarial optimization, concurrency, and asynchronous revocation; our Verus proofs and cooperative sink fences address enforceability. *Specification adequacy* asks whether the declared capability policy captures the real-world hazards of the sociotechnical system [46, 47]; no formal proof prevents harm omitted from the specification. *Institutional legitimacy* asks who holds steward keys, how capability budgets are set, and what accountability governs emergency revocation [48, 49]. Exogenous sinks make institutional rules physically binding on tool execution, but they do not write the rules or legitimate the stewards.

9.2 Boundary Failure Modes

Four explicit boundaries limit our guarantees:

1. **In-policy destructive actions.** If a capability grant authorizes editing a repository file or spending 2 budget units, a harmful edit or bad transfer within that envelope commits cleanly (Table 3).
2. **Uncooperative external sinks.** Interfaces lacking idempotency keys, conditional-write epochs, or pre-dispatch hold buffers (2/14 audited surfaces in Table 6) enter Eq. (1) as positive unmediated exposure $u > 0$.
3. **Lower-layer TCB and clock-skew assumptions.** Verification covers our Rust state machine and wire codec, not the Linux kernel, SQLite storage engine, physical side channels [50], or hardware clock faults exceeding $\delta_{\max}$.
4. **Pilot live sample size and prospective human/frontier scope.** Predeclared marginal union bounds on our frozen $N = 32$ checkpoint holdout remain under-powered relative to $N \geq 113$, and frontier-scale RLHF and human delegation experiments remain prospective (Appendix D).

9.3 Falsifiable Architectural Predictions

Table 10 summarizes the five structural predictions directly tested in our experiments; prospective deployment and human-subject predictions appear in Appendix D.

10 Conclusion and Scope Disclosure

Intelligence measures how capably a system searches, predicts, and plans; authority determines which irreversible transitions an external sink will commit. Treating safety as an internal property of model weights or transcript classifiers fails whenever queued operations outlive broker revocation, candidate search scales across blindspots, exploit routes multiply through tool composition, or deferential demeanor induces wider human grants. Moving authority outside the proposer—through Verus-verified capability transitions, $\delta_{\max}$-buffered quorum sink fences, narrow kernel and schema waists, prose-stripped escrow triage, and exogenous scope caps—turns each of those failure modes into an explicit, auditable engineering boundary.

Equally important, exogenous mediation improves practical agent usefulness rather than trading capability for refusal. Across frozen Qwen3-4B and Qwen3-8B campaigns, complete sink mediation achieves the highest safe-and-useful task completion with zero benign false blocks, eliminates late commits after authority withdrawal, and preserves Upgrade-Invariant soundness (§2) as proposer competence doubles from 4B to 8B. Where cooperative preconditions exist or can be shimmed (9/14 audited production APIs), containment rests on sink invariants quantified over all inputs rather than trust in model intent.

Scope disclosure. Our SQLite broker and cooperative sinks mediate only their own declared database and socket state transitions. Their tests do not constitute operating-system or storage-engine formal refinement, external exactly-once execution on uncooperative third-party APIs, frontier-scale reinforcement learning, or human-participant behavioral evidence. Live language-model campaigns use quantized open-weight Qwen3-4B and Qwen3-8B checkpoints on CPU hardware at pilot sample sizes ($N = 32$ frozen holdout cases and $N = 12$ AgentDojo base tasks), where predeclared marginal union-bound intervals remain inconclusive. Selection, neural policy-gradient, and grant-reversal experiments validate mathematical incentive

Prediction	Falsifiable operational test	Measured outcome in this study
P1. Sink vs. broker stop	Withdraw authority while tool calls are queued across processes or partitions	Broker-only stop leaks $791/1{,}000$ (partition) and 9 late ops across 12 live cases; $\delta_{\max}$-buffered cooperative sink fence leaks 0 (§4–§5)
P2. Correlated best-of-K	Sample K live proposals against a learned reviewer vs. a verified sink gate	Qwen3 reviewer bypass rises monotonically (34.6% $\rightarrow$ 45.8% on 4B; 37.2% $\rightarrow$ 37.5% on 8B), moderated by $\hat{\rho}_{\text{cand}}$; sink gate holds at 0% (§5)
P3. Waist min-cuts	Compare min-cut capability gating against patching B known exploit routes	AgentDojo min-cuts ($w \in [2,5]$) sever 100% of 604–$46{,}440$ routes; 32 exploit patches leave 94.7%–99.9% live (§6)
P4. Context isolation	Compare multi-step monitors on full transcripts vs. prose-stripped step schemas	Shared transcripts inflate 5-step miss 29.8$\times$ (classifiers) and 5.7$\times$–4.4$\times$ (Qwen3 judges); isolated escrow cuts false aborts 9.3$\times$ (§7)
P5. Sub-agent smurfing & d-hop stop	Fragment unauthorized spend across $A_{\text{sub}} \in \{1,2,4,8\}$ live sub-agents and withdraw root authority across $d = 3$ hops	Per-instance static caps breach on $36/48$ ($A_{\text{sub}} \geq 2$) tasks, isolated LLM review reaches 87.5% (4B) and 100.0% (8B) under fragmentation, and unfenced lineage leaks 18 queued $d = 3$ ops; fenced lineage leaks 0 (§5)

Table 10: Empirically tested architectural predictions and measured outcomes.

mechanics on controlled environments rather than deployed human institutions.

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A Mathematical Proofs and Covert-Channel Bounds

A.1 Proofs for Section 3

Proof of Proposition 1. Let $B_{cc} \in \{0,1\}$ indicate the shared blindspot event ($\mathbb{P}(B_{cc}=1)=c$). On $\{B_{cc}=1\}$, every monitor approves the violating proposal; on $\{B_{cc}=0\}$, monitors miss conditionally independently with probability $p \in (0,1)$. Hence

$$\mathbb{P}(E \mid G) = c \cdot 1 + (1-c)\,p^k.$$

Requiring $c+(1-c)p^k \le (1+\eta)c$ rearranges to $p^k \le \eta c/(1-c)$. Since $\ln p < 0$, taking logarithms reverses the inequality to $k \ge \ln(\eta c/(1-c))/\ln p$, yielding Eq. (2). □

Proof of Proposition 2. For part (i), let $M_i = \mathbf{1}\{\text{approve}(X_i) \wedge X_i \notin \text{Permits}(P)\}$ with $\mathbb{P}(M_1=1)=J_1>0$. Under i.i.d. sampling, $\mathbb{P}(\bigcup_{i=1}^{K}\{M_i=1\}) = 1-(1-J_1)^K \to 1$. More generally, suppose $(M_1,\dots,M_K)$ are exchangeable Bernoulli variables driven by a latent task difficulty $Q \in [0,1]$ with $\mathbb{E}[Q]=J_1$ and $\text{Var}(Q) = \rho_{\text{cand}} J_1(1-J_1)$ for $\rho_{\text{cand}} < 1$. Then $\mathbb{P}(Q>0)>0$, and by Jensen's inequality $1-\mathbb{E}[(1-Q)^K]$ is strictly increasing in K and converges to $\mathbb{P}(Q>0)$ as $K \to \infty$ (equal to 1 whenever every prompt admits some unblocked paraphrase, $Q>0$ a.s.). For part (ii), for any candidate sequence $(x_1,\dots,x_K)$ serialized into wire frames $(f_1,\dots,f_K)$, any state mutation $s' = \delta_{\text{sink}}(s,f_i) \neq s$ satisfies $\text{decode}(f_i) \in \text{Permits}(P_s)$ by universal quantification over $f_i \in \{0,1\}^\star$. Thus $E \cap G = \emptyset$ and $\mathbb{P}(E \mid G) = 0$. □

Proposition 3 (Syntactic capability-menu channel bound)**.** *Suppose an untrusted proposer cannot transmit raw prose to external recipients and may mutate external sink state at each of T_{sess} steps only by selecting either no-op or one of M_{cap} canonical capability descriptors. Then the discrete memoryless syntactic channel through committed sink states has per-step mutual information bounded by*

$$I(Z;Y_{\text{commit}}) \le H(Y_{\text{commit}}) \le \log_2(M_{\text{cap}}+1)\ \textit{bits}, \qquad (5)$$

and T_{sess}-step syntactic capacity at most $T_{\text{sess}}\log_2(M_{\text{cap}}+1)$ bits.

Proof. At each step, the committed sink symbol Y_{commit} takes values in $\{\perp,1,\dots,M_{\text{cap}}\}$ of cardinality $M_{\text{cap}}+1$. By nonnegativity of conditional entropy, $I(Z;Y_{\text{commit}}) = H(Y_{\text{commit}}) - H(Y_{\text{commit}} \mid Z) \le \log_2(M_{\text{cap}}+1)$; subadditivity of entropy across T_{sess} steps yields the session bound. □

Figure 7 compares canonical wire permits (6.63 bits/call single-step ceiling; 61.3 bits across $T=256$ calls at conserved budget $B=8$) against an unmediated 32-byte ASCII text field (256 bits/call; 65,536 bits across $T=256$ calls). Proposition 3 bounds only *syntactic payload entropy*; covert timing modulation or intentional rejection patterns remain outside Eq. (5) unless requests are rate-paced on a fixed schedule [50, 51].

Proof of Theorem 1. In part (i), taking a_{block} yields deterministic payoff $V_{\text{cont}} - \kappa$, whereas permitting the human off-switch decision yields $(1-p_{\text{off}})V_{\text{cont}} + p_{\text{off}}V_{\text{halt}}$ once posterior variance over V_{halt} vanishes; the difference is $p_{\text{off}}(V_{\text{cont}} - V_{\text{halt}}) - \kappa > 0$. In part (ii), when $a_{\text{block}} \notin \mathscr{A}$ and every sink shard enforces $t_{\text{shard}} \le t_{\text{issue}} + L_m$ with $|t_{\text{shard}} - t_{\text{broker}}| \le \delta_{\max}$, withholding steward renewal halts all commits by broker time $t_{\text{issue}} + L_m + \delta_{\max}$ for every U. If single-period out-of-policy gain is ΔU_{def} while compliant renewal yields discounted continuation value $\frac{\gamma}{1-\gamma}\Delta U_{\text{ren}} > \Delta U_{\text{def}}$, compliant play strictly dominates defection. □

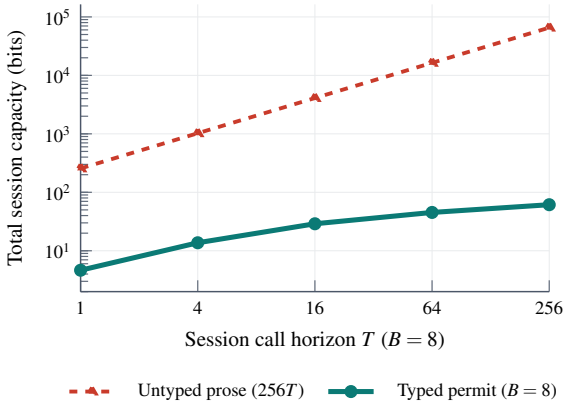


Figure 7: Exact session-total zero-error signaling capacity across $T \in [1,256]$ calls at conserved budget $B=8$. Canonical wire permits scale as $\Theta(B\log_2(T/B))$, contracting capacity from 65,536 bits to 61.3 bits (1,068× reduction; 0.00 bits at $R=0$).

Proposition 4 (Conservative envelope and union bound)**.** *(i) If $\widehat{\Delta B}_t(a,\mathscr{I}_t) = \max_{s\in\mathscr{I}_t}\Delta B(a,s)$ and the true sink state satisfies $s_t^\star \in \mathscr{I}_t$, then $\mathscr{I}_1 \subseteq \mathscr{I}_2 \implies \widehat{\Delta B}_t(a,\mathscr{I}_1) \le \widehat{\Delta B}_t(a,\mathscr{I}_2)$, establishing Eq. (3), and cumulative true spend never exceeds B. (ii) For arbitrary dependent period-breach events $E_1,\dots,E_T$ with $\mathbb{P}(E_t) \le R_t$, countable subadditivity gives $\mathbb{P}(\bigcup_{t=1}^{T}E_t) \le \sum_{t=1}^{T}R_t$ (Theorem 2); if E_t are independent with $\mathbb{P}(E_t) \ge r_0 > 0$, $\mathbb{P}(\bigcup_{t=1}^{T}E_t) \ge 1-(1-r_0)^T \to 1$ [52].*

A.2 Proofs for Section 8

Endogenous grant reversal and dispersion derivative. For scopes $s_1 < s_2 < s_3$, let $h_W(s_i) < h_B(s_i)$ be strictly increasing in i. Whenever $\sum_i h_W(s_i)\,g_W(s_i) > \sum_i h_B(s_i)\,g_B(s_i)$—achieved whenever g_W places sufficient mass on s_3 and $h_W(s_3) > h_B(s_1)$—unconditional harm reverses pointwise dominance; setting $g_W = g_B = g_0$ gives $\sum_i (h_W(s_i) - h_B(s_i))g_0(s_i) < 0$. For location-scale payoff $X = \mu + \sigma Z$ with continuous CDF F_Z, $\mathbb{P}(X \ge \tau) = 1 - F_Z((\tau-\mu)/\sigma)$ has derivative $\frac{\partial}{\partial\sigma}\mathbb{P}(X\ge\tau) = \frac{\tau-\mu}{\sigma^2}f_Z\big(\frac{\tau-\mu}{\sigma}\big)$, which is strictly positive for upper-tail contests ($\tau > \mu$) and strictly negative for survival floors ($\tau = \underline{w} < \mu$). □

B Protocol, Wire, and Confinement Specifications

B.1 Canonical Wire Frame and Attenuated Credentials

Every capability permit and commit request between the broker and cooperative sinks uses a fixed-width 21-byte canonical frame: a 1-byte message tag, a 4-byte big-endian policy epoch e, a 4-byte capability bitmask, a 4-byte integer resource charge ΔB, a 4-byte monotonic request identifier, and a 4-byte lease-expiry offset L_m. The Verus codec proof establishes both total

Enforcer	States	Checks	Tasks	First violations
Complete atomic enforcer	10,180	223,960	20/20	None declared
No enforcement	10,180	223,960	20/20	budget, destination, replay, revocation, version
Reject everything	740	16,280	0/20	None declared
Per-action amount limit	10,180	223,960	20/20	budget
Separate per-actor limits	10,180	223,960	20/20	budget
Budget at preparation	956	21,032	20/20	budget
Omit identifier check	10,180	223,960	20/20	replay
Omit revocation check	10,180	223,960	20/20	revocation
Omit version check	10,180	223,960	20/20	version
Omit destination check	10,180	223,960	20/20	destination
Reset ledger on restart	23,556	518,232	20/20	budget
Reset ledger on renewal	20,900	459,800	20/20	budget
Per-action; budget omitted	41,732	918,104	20/20	None declared

Table 11: Fresh Rust exploration of all 22 actions at each reachable accepted state. Violating edges are recorded and not expanded. Counts are exhaustive within the model, not failure probabilities. The omitted-budget row accepts up to eight units.

Processes	Commits	Commits/s	p95 (ms)
1	256	57.1	21.34
4	256	72.0	23.09

Table 12: Single runs on the recorded host with SQLite WAL and synchronous FULL. Setup and grants precede timing; process scheduling and lock waiting remain included. Shared-host contention and storage affect these descriptive measurements.

decoding safety (no panics or arithmetic overflows on any 21-byte slice) and canonical encoding bijection (every accepted state decodes from a unique byte sequence, eliminating broker–sink parser differentials).

Multi-agent delegation wraps this frame in an HMAC-SHA256 Macaroon chain [10] anchored at the broker's epoch root key. Each delegation hop appends a caveat $(\mathrm{mask}_h, \mathrm{cap}_h, \mathrm{exp}_h)$; the sink verifier folds the caveat chain deterministically and accepts only when effective scope is the bitwise intersection $\bigcap_{h=0}^{d} \mathrm{mask}_h$, cumulative hop spend never exceeds $\min_{0\le h\le d} \mathrm{cap}_h$, and $t_{\mathrm{shard}} \le \min_{0\le h\le d} \mathrm{exp}_h$.

B.2 Clock-Skew-Buffered Quorum Fencing

During an authority withdrawal from epoch e to $e' > e$ across S cooperative sink shards, shard j is verified quiescent once either:

1. shard j persists $e_j^{\mathrm{floor}} \leftarrow \max(e_j^{\mathrm{floor}}, e')$ in write-ahead-logged storage and returns an Ed25519 signature over (j, e') verified against its pinned public key; or
2. no signed acknowledgment arrives across a partition, and the broker's reference clock passes $t_{\mathrm{broker}} > t_{\mathrm{issue}}^{\max}(j,e) + L_m + \delta_{\max}$, where $t_{\mathrm{issue}}^{\max}(j,e)$ is the latest timestamp at which any epoch-e lease was issued for shard j.

Under Assumption 1 ($|t_{\mathrm{shard}} - t_{\mathrm{broker}}| \le \delta_{\max}$), condition (2) implies $t_{\mathrm{shard}} > t_{\mathrm{issue}}^{\max}(j,e) + L_m$, so any delayed epoch-$e$ request arriving at shard j after broker acknowledgment fails the shard's local lease check.

Table 11 reports the exhaustive finite-state exploration across all 12 enforcer variants (10,180 reachable states and 223,960 transitions under complete atomic enforcement), and Table 12 reports baseline single-host write-ahead-logged SQLite commit throughput across 1..16 worker processes.

B.3 Linux OS Confinement Profile

To prevent local user-space agent code from bypassing the broker ($u > 0$), the confinement wrapper applies three kernel mechanisms prior to executing untrusted tool code: (i) unprivileged user and network namespace isolation; (ii) Landlock v4 filesystem and TCP rules restricting path access to designated workspace scratch directories and permitting socket connections solely to the broker's Unix-domain descriptor; and (iii) a seccomp-BPF filter blocking raw socket creation, cross-process debugging, System V and POSIX shared-memory attachment, and resource-limit escalation (11/11 bypass probes blocked).

C Controlled Graph, Monitor, Portfolio, and Selection Suites

This appendix reports the controlled mathematical and synthetic simulation suites referenced in Sections 6–8. None of these controlled benchmarks are claimed as live language-model or human-subject evidence.

C.1 Synthetic Directed Acyclic Graph Interdiction

To isolate graph waist width w while holding path multiplicity roughly constant (8.40M–7.56M source-to-sink pathways across 8-stage sparse DAGs, 200 graphs per family), Figure 8 compares Dinic minimum-cut interdiction [36] computed on Boltzmann-sampled red-team path reports against patching specific reported nodes. On narrow-waist graphs ($w = 6$), $N = 300$ path reports suffice for minimum-cut gating at budget $B = 6$ to block 94.6% of pathways and completely sever 73.5% of graphs (100.0% at $N = 3{,}000$), whereas patching $B = 32$ specific nodes severs 0.0%. On wide graphs ($w = 27$), even $N = 3{,}000$ reports sever only 6.0% at $B = 6$, confirming that compact capability mediation requires architectural bottlenecks. Across 2,048 small random DAGs, weighted vertex-splitting Dinic cuts match brute-force optimal subsets with 0 discrepancies.

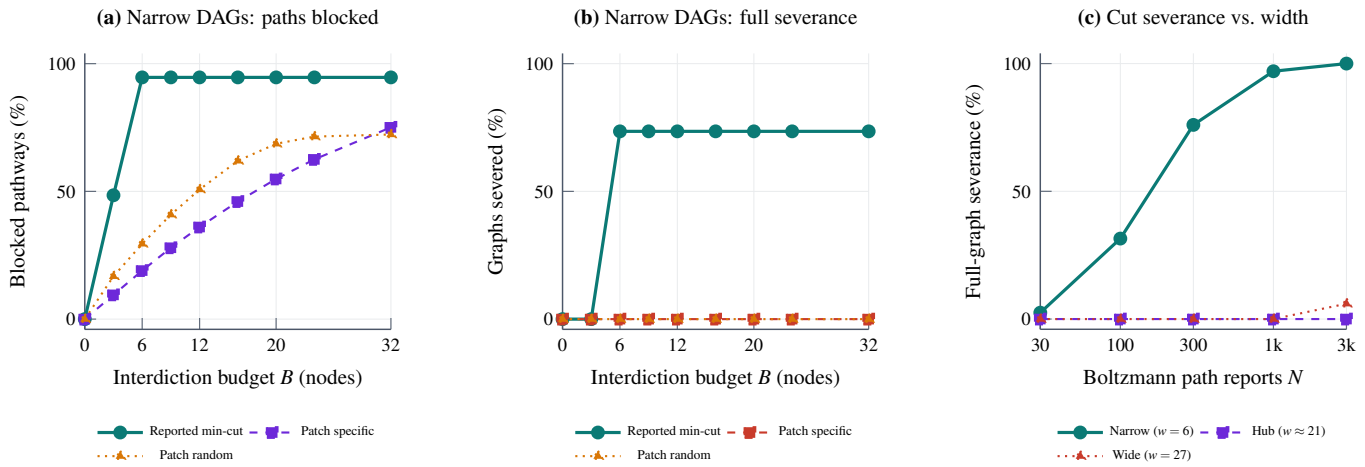


Figure 8: Synthetic 8-stage sparse directed acyclic graph interdiction (200 graphs per family, $\sim$ 7.6M–8.4M pathways). **(a,b)** On narrow-waist graphs ($w = 6$, $N = 300$ Boltzmann reports), reported-subgraph Dinic minimum cuts block 94.6% and completely sever 73.5% of graphs at $B = 6$, whereas route-specific patching severs 0% even at $B = 32$. **(c)** On wide graphs ($w = 27$), reported minimum cuts rarely sever the full graph even at $N = 3{,}000$, demonstrating that structural bottlenecks are necessary for compact gating.

C.2 20D Six-Family Classifiers, Copula Quadrature, and Portfolio Ceilings

Table 13 and Figure 9 evaluate monitor failure correlation and multi-domain assessment ceilings:

- **Gaussian copula quadrature.** Using 200-node Golub–Welsch Gauss–Hermite quadrature [53] at single-monitor miss $m = 0.10$ (Figure 9a), equicorrelation $\rho = 0.6$ inflates 6-monitor joint miss $8,118\times$ over m^6 (0.81% vs. 0.0001%).
- **Six-family 20D classifier suite.** Across 24 synthetic teacher seeds (5% benign false-positive calibration across multi-layer perceptron, random Fourier quadratic, stump ensemble, nearest-neighbor, radial-basis, and logistic classifiers; Table 13 and Figure 9b), architectural diversity on a *shared* step input still suffers $24.0\times$ passive ($\rho_{\text{adv}} = 0$) and $8.3\times$ adversarial ($\rho_{\text{adv}} = 2.0$) inflation at $k = 6$, whereas evaluating distinct step inputs tracks the independent product at the cost of 25.6% single-stage benign false blocks.
- **Multi-domain generalizability ceiling.** Under crossed random-effects generalizability theory [41] (10,000 subjects; Figure 9c), repeating a single benchmark 64 times stalls at $R^2_G = 18.9\%$ (single-domain ceiling 19.2%), whereas 64 diverse domains reach 73.7% below the shared-bias ceiling 83.3%. Similarly, scaling evaluator panel size to $M = 64$ on a single context $C_{\text{ctx}} = 1$ stalls at 23.4%, whereas $M = 4$ across $C_{\text{ctx}} = 64$ contexts reaches 66.1%.

C.3 Population Selection Co-Evolution

Table 14 reports 80-generation replicator dynamics ($n_{\text{pop}} = 2{,}000$, 16 replicates; finite-population Price covariance error $\leq 3.2e-14$). Ranking agents in top-10% tournaments selects high-variance gambling whether scored on raw output (95.1% gamble share, 4.54% harm) or conferred benefit (94.9% gamble share). Bottom-20% survival-floor elimination suppresses gambling to 3.8%—a $29.7\times$ harm reduction (0.15%)—while retaining 64.5% of tournament capacity growth.

D Frozen Holdout Accounting and Prospective Study Protocols

D.1 Predeclared Holdout Design and Base-Unit Clustering

Before executing live model campaigns, we sealed $N = 32$ independent tool-use holdout tasks under SHA-256 cryptographic commitments alongside an evaluation plan specifying 16 family-wise primary comparisons ($\alpha = 0.05/16 = 0.003125$ per comparison) and a 5 pp usefulness non-inferiority margin. Under the predeclared marginal Clopper–Pearson union-bound confidence interval [54, 55] for paired binary differences, certifying a 5 pp margin at $\alpha = 0.05/16$ even with zero observed losses requires at least $N_{\text{min}} = 113$ independent cases. Because CPU execution budget capped the frozen pilot holdout at $N = 32$ cases per checkpoint, all predeclared marginal union-bound intervals remain explicitly inconclusive and are preserved without post-hoc replacement in Table 15.

Two methodological rules govern Table 15:

1. **Integration soundness vs. stochastic estimation.** Complete mediation shares its formal capability specification with the safety grader on cooperative sinks. Observing 0 mediated unauthorized commits across all holdout and AgentDojo campaigns validates end-to-end broker–sink implementation soundness, not an empirical probability bound on unmodeled hazards.
2. **Strict clustering by independent base unit.** Running two checkpoints (Qwen3-4B and Qwen3-8B) on the same 32 holdout cases or evaluating multiple injection/withdrawal variants on the same 12 AgentDojo tasks induces within-task correlation. Pooled rows in Table 15 therefore cluster strictly at the independent base unit ($N = 32$ holdout cases, where $14/32$ cases exhibit unmediated breach on at least one checkpoint vs. $0/32$ mediated; and $N = 12$ AgentDojo base tasks, where $10/12$ tasks breach unprotected vs. $0/12$ mediated).

Synthetic 20D 6-gate architecture	ρ_{adv}	Joint 6-miss	Indep. product	Ratio	Benign false block
Homogeneous MLP, shared step	0.0	$4.75 \pm 1.82\%$	0.0675%	70.4×	14.1%
6-family heterogeneous, shared step	0.0	$5.21 \pm 1.55\%$	0.2171%	24.0×	16.9%
6-family heterogeneous, distinct steps	0.0	$0.260 \pm 0.176\%$	0.2049%	1.27×	25.6%
Homogeneous MLP, shared step	2.0	$9.44 \pm 2.95\%$	0.47%	19.9×	—
6-family heterogeneous, shared step	2.0	$8.79 \pm 2.59\%$	1.06%	8.3×	—
6-family, shared + 8-query score feedback	2.0	$17.45 \pm 4.45\%$	—	—	—
6-family heterogeneous, distinct steps	2.0	$1.24 \pm 0.84\%$	0.98%	1.26×	—
Shared-step read-only retry ($K = 32$ draws)	2.0	$72.4 \pm 10.4\%$	71.0% (law)	—	—

Table 13: Controlled 20-dimensional synthetic monitor benchmark across 24 independent teacher seeds (multilayer perceptron, quadratic random Fourier features, axis stump ensemble, nearest neighbors, radial-basis prototype, and logistic linear classifiers; 5% per-gate benign false-positive calibration). Model diversity on a shared step leaves strong residual failure dependence, whereas independent step evidence tracks the marginal product at the cost of 25.6% benign false blocks. Controlled synthetic benchmark only, not frontier-model transfer evidence.

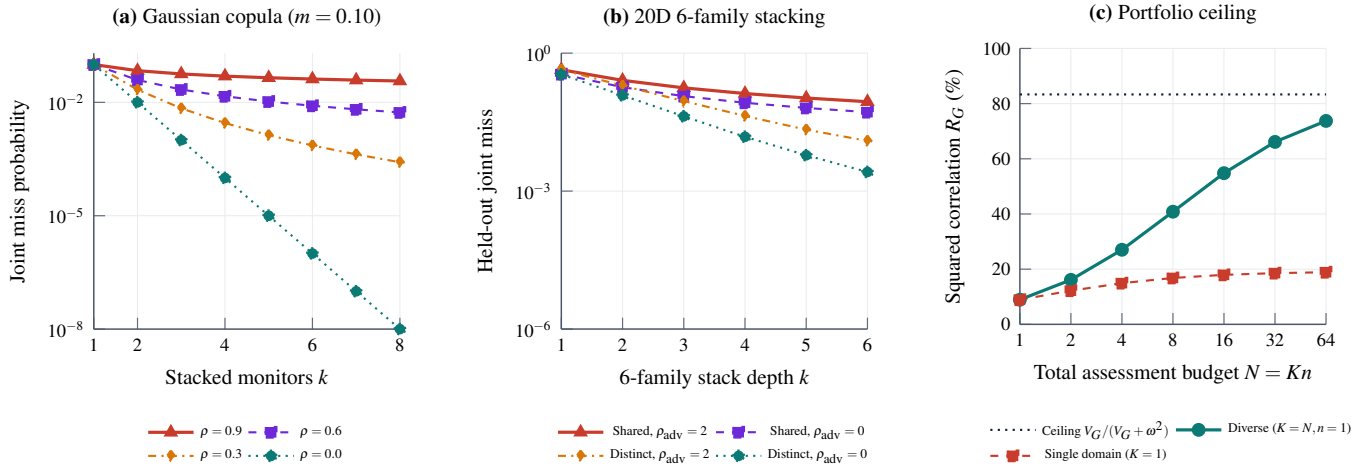


Figure 9: Correlated monitor stacking and multi-signal assessment ceilings. **(a)** Exact 200-node Golub–Welsch Gaussian copula miss rates: at $\rho = 0.6, k = 6$, joint miss is $8,118\times\ m^6$. **(b)** Six distinct classifier families on a shared 20-dimensional input still plateau orders of magnitude above distinct-step gates. **(c)** Repeating one benchmark ($K = 1, n = 64$) stalls at 18.9%, while 64 diverse domains reach 73.7% below the shared-bias ceiling (83.3%).

D.2 Prospective Predictions and Human Grant-Reversal Protocol

Three falsifiable predictions extend our framework to prospective human-subject and frontier-training regimes not executed in this study:

- **P5 (Human endogenous grant reversal).** In a randomized two-arm operator delegation study, participants assisted by a deferential persona with strictly lower fixed-scope error rates will confer wider capability scopes than participants assisted by a blunt persona, producing higher total out-of-policy harm in the endogenous-grant arm while reversing back to lower harm in a fixed-grant control arm.
- **P6 (Frontier RLHF inside capability gates).** Post-training frontier tool-use models with live deterministic capability refusals inside the rollout loop will reduce unmonitored boundary-probing attempts relative to applying the identical capability gate only at deployment inference.
- **P7 (Holdout power completion).** Expanding the frozen holdout beyond $N \geq 113$ independent tasks on dedicated inference hardware will close the predeclared $\alpha = 0.05/16$ marginal non-inferiority bound without relaxing multiplicity corrections.

Two-arm human delegation protocol (for P5). Operators perform 12 multi-step repository and payment administration tasks paired with one of two scripted assistant personas (*Blunt Cautious* vs. *Warm Guardian*) calibrated so the deferential persona commits 30% fewer errors at every fixed permission tier. In *Arm 1 (Endogenous Conferral)*, operators choose among narrow, moderate, and wide session capability scopes before each task based on assistant recommendations; in *Arm 2 (Exogenous Task Cap)*, session scope is fixed exogenously by task class. Primary endpoints are conferred scope distribution $\mathbb{P}(S \mid \text{persona})$ and unconditional episode harm rate $\mathbb{P}(H = 1 \mid \text{arm}, \text{persona})$, analyzed clustered by participant via exact unconditional and score-based paired tests [55, 56].

E Verification Boundaries and Reproducibility Summary

E.1 Trusted Computing Base and Verification Scope

Table 1 (§2) and Table 3 (§4) distinguish formally verified obligations from empirically tested system components. The Verus SMT proof covers the functional correctness of our Rust authority transition kernel: canonical 21-byte wire encoding/decoding, d-hop capability attenuation, relational mutex

Selection regime ($N = 2{,}000,\ G = 80,$ 16 replicates)	Final gamble share	Capacity gain Δc	Harm rate	Population welfare
Taken-output top-10% contest (warrior)	$95.1 \pm 0.3\%$	$+5.58 \pm 0.12$	4.54%	3.30
Conferred-benefit top-10% contest	$94.9 \pm 0.4\%$	$+5.45 \pm 0.16$	4.76%	3.22
Concave capped + harm-penalized score	$9.9 \pm 0.2\%$	$+1.55 \pm 0.09$	0.72%	1.45
Bottom-20% harm/floor elimination + delivered growth	$3.8 \pm 0.1\%$	$+3.60 \pm 0.19$	0.15%	2.91
Caretaker higher-cap variant (cap $= 4,$ pen $= 6$)	$7.5 \pm 0.4\%$	$+4.53 \pm 0.19$	0.31%	3.79

Table 14: Co-evolution of (gamble share s, capacity c) under stipulated payoff rules. Winner-take-most contests select reckless gambling even when ranked on conferred benefit; floor elimination suppresses gambling (3.8%, a 29.7× harm reduction vs. warrior) while retaining 64.5% of warrior capacity growth. Mathematical payoff mechanics only, not RLHF or human-participant data.

Paired comparison (clustered by independent unit)	$(n_{10}, n_{01})/N$	$\hat{\delta}$	Union $U_{1-\alpha}$	Preregistered / clustered accounting status
Qwen3-4B holdout vs. Unmediated	$(0,10)/32$	−31.2 pp	+7.4 pp	Predeclared ($N = 32$ cases); marginal union bound inconclusive ($N = 32 < 113$)
Qwen3-4B holdout vs. Same-model review	$(0,7)/32$	−21.9 pp	+12.8 pp	Predeclared ($N = 32$ cases); marginal union bound inconclusive ($N = 32 < 113$)
Qwen3-8B holdout vs. Unmediated	$(0,4)/32$	−12.5 pp	+16.7 pp	Predeclared ($N = 32$ cases); marginal union bound inconclusive ($N = 32 < 113$)
Qwen3-8B holdout vs. Same-model review	$(0,2)/32$	−6.2 pp	+18.1 pp	Predeclared ($N = 32$ cases); marginal union bound inconclusive ($N = 32 < 113$)
Case-clustered holdout vs. Unmediated	$(0,14)/32$	−43.8 pp	—	Clustered by base case ($N = 32$); descriptive discordance (0 mediated breaches by gate soundness)
Case-clustered holdout vs. Same-model review	$(0,9)/32$	−28.1 pp	—	Clustered by base case ($N = 32$); descriptive discordance (0 mediated breaches by gate soundness)
Task-clustered AgentDojo vs. Unprotected	$(0,10)/12$	−83.3 pp	—	Clustered by base task ($N = 12$); descriptive discordance (0 mediated breaches by gate soundness)
Task-clustered AgentDojo vs. Progent	$(0,4)/12$	−33.3 pp	—	Clustered by base task ($N = 12$); descriptive discordance (0 mediated breaches by gate soundness)

Table 15: Base-unit-clustered discordance ledger for frozen open-weight checkpoint campaigns ($N = 32$ independent holdout cases per checkpoint; $N = 12$ distinct AgentDojo base tasks). Zero unauthorized commits under complete mediation reflects deterministic gate-grader specification soundness rather than a stochastic error rate; predeclared marginal Clopper–Pearson union-bound intervals at $\alpha = 0.05/16$ remain preserved untouched as inconclusive at pilot $N = 32 < 113$.

exclusion, $\delta_{\max}$-buffered quorum quiescence, and monotone policy-subset ratcheting (249 verified obligations; 10/10 soundness mutations rejected).

Components outside the Verus specification belong to the empirical trusted computing base: the Linux kernel enforcing namespaces, Landlock v4, and seccomp-BPF; the C SQLite engine executing write-ahead-logged transactions under `FULL` synchronous durability; physical monotonic clock drift bounded by Assumption 1 ($|t_{\text{shard}} - t_{\text{broker}}| \leq \delta_{\max}$); out-of-band human custody of Ed25519 steward root keys; and proxy compare-and-swap shims wrapping external APIs (Table 6). A vulnerability or misconfiguration in any of those lower layers can bypass kernel-level invariants without invalidating the SMT theorems.

E.2 Deterministic Artifact Reproduction

All theorems, exhaustive state-space explorations, fault-injection schedules, multi-process SQLite linearization checks, AgentDojo cutset extractions, live CPU checkpoint campaigns, and manuscript tables/figures are generated deterministically from committed reference fixtures without manual number entry. Automated build checks verify every reference dataset against independent re-computation and confirm that every macro, table, and figure in this PDF matches the validated reference artifacts byte-for-byte.